

The Assertion Ladder: Graduated Assertion for Embedding a Technical Domain in Repositories Shared with Coding Agents

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ABSTRACT Coding agents now contribute to repositories alongside people. They write quickly, they start every session with no memory of the last one, and their changes arrive faster than a person can review them closely. Two forces follow: *sprawl*, in which locally plausible changes spread one domain rule across many places, and *cognitive erosion*, in which neither the reviewer nor the agent can hold enough of the domain in mind to notice. Current practice answers with architecture checks, executable specifications and instruction files for agents, but it has no standard way to combine them: no shared vocabulary for how strongly a rule is asserted, no link from agent instructions to enforcement, no closure between requirements and scenarios, and no benchmark of whether a repository's own harness catches the mistakes that matter. We propose the *Assertion Ladder*, a pattern whose six rungs run from a rule that is merely implied, through one that is shown, told, checked and explained, to one that cannot be written wrongly at all, realised by nine smaller patterns. Read through lean thinking, the ladder is standard work, with its tests built in, for a contributor who cannot remember. We show it in two settings. A greenfield service, presented in TypeScript with equivalent Java and Python implementations, binds every rung; seeding fourteen canonical mistakes into each implementation, its single verify command catches all 42, where before the requirement and behavioural rungs it caught 30 of 36. A production TypeScript platform, Tutors, shows the ladder retrofitted while agents were already contributing, and its release checks extracted into a reusable harness. The repositories are public.

KEYWORDS Coding agents, Software architecture, Architecture testing, Behaviour-driven development, EARS, Lean software development, Design patterns, Human–AI collaboration.

1. Introduction

A new kind of contributor has joined software teams. Coding agents, language models given a shell, a file system and a test runner, now read a repository, plan a change, edit many files and run the tests before handing the result to a person (M. Chen et al. 2021; Jimenez et al. 2024; Yang et al. 2024). They are fast

and broadly competent. They are also, in one respect, the most difficult colleague a team has ever had: every session starts with no memory of the last one. Whatever the agent learned about the domain yesterday, about why the work-in-progress limit is checked in the service rather than the route, or why the domain objects never import the web framework, is gone this morning.

The only thing a person and an agent reliably share is the repository. Whatever the repository does not say, and does not enforce, the agent will reinvent, usually plausibly and often differently each time. People have the opposite problem. They remember the domain, but they cannot read every line an agent writes; studies of programmers working with code-generating tools describe a shift from writing to reviewing, with the attendant risk that review becomes skimming (Vaithilingam et al.

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2022; Barke et al. 2023). Security studies show what can pass unnoticed (Pearce et al. 2022; N. Perry et al. 2023).

We name the two resulting forces. *Sprawl* is what happens when a large volume of locally plausible changes spreads one domain rule across many places: a check is copied into a route because it was nearby, a notification is called inline because that was quicker than publishing an event, a framework annotation is placed on a domain object because it was convenient. Each change is defensible; together they dissolve the domain. *Cognitive erosion* is what happens when neither party can hold enough of the domain in mind to notice: the reviewer checks syntax because the intent is not visible in the diff, and the agent re-derives conventions from scratch every session. Lehman observed long ago that a system’s structure decays under change unless work is done to preserve it (Lehman 1980). Agents do not change that law; they change its rate.

The need for constraints A common response is to write more guidance: longer prompts, longer contributor guides, longer instruction files for agents (*AGENTS.md: A Simple, Open Format for Guiding Coding Agents* n.d.). Guidance helps, but it has two failure modes. It is read at the start of a session and then competes with everything else in the agent’s context, and it grows until nobody, person or agent, reads it at all. Our earlier work proposed a repository standard in which conventions are carried three ways at once: exemplified by one small vertical slice, declared in a manifest of a few hundred words, and enforced by architecture rules written as tests (Griffin et al. 2026). This paper generalises that idea. It asks not only *whether* a rule is written down, but *how strongly* it is asserted, and it extends the question from structure to behaviour and requirements.

The Assertion Ladder Our central claim is pedagogical. A coding agent is a capable learner with no memory; the repository is its teacher. Teachers do not assert everything the same way. They show a worked example, they tell a rule, they set an exercise that marks the answer, they give feedback that explains the mistake, and occasionally they arrange the task so that the wrong answer is simply not available. The *Assertion Ladder* (Figure 1) gives each of these moves a rung, from **L0 Implied** to **L5 Prevented**, and asks the designer of a repository to place every domain rule on a rung deliberately: high enough that the learner cannot miss it, and no higher than the rule’s risk justifies, because every assertion also costs something to write, to maintain and to read.

Lean as the explanation The ladder is not new engineering so much as old engineering applied to a new worker. Lean thinking starts from *value*, what the customer and here also the domain need, and treats everything that does not serve it as waste (Womack & Jones 1996; Poppendieck & Poppendieck 2006). For work that repeats, its central tool is *standard work*: the current best method, followed by everyone who does the job and improved by the team that owns it (Liker 2004). It reads a process as a *value stream* and asks where along it a defect is found and how much waiting and rework that costs (Rother & Shook 1999). And it answers each problem with *root cause analysis*, asking why until the answer is something the process

can change (Ohno 1988). Read this way, the rungs are standard work for a contributor who cannot remember it: shown in a worked slice, told in a short manifest, and then built into the work as checks that stop the build, explain the stop, or make the mistake impossible to write. Section 8 argues that lean explains why the ladder works rather than merely decorating it.

Critical gaps Software engineering already has most of the parts. Architecture conformance checking, specification by example, structured requirements and instruction files for agents all exist, and Section 9 reviews them. What it lacks is a standard way to combine them. We identify five gaps in current practice that call for standardisation: there is no shared vocabulary for how strongly a rule is asserted (G1); instruction files for agents are unvalidated prose with no link to enforcement (G2); conformance tools say nothing about which rules to check rather than document, or that a failure should name its fix (G3); agent-facing repositories have no executable closure between requirements and the scenarios that test them (G4); and there is no benchmark of whether a repository’s own harness catches the mistakes that matter, as opposed to whether an agent can complete a task (G5). Each gap is small; together they mean that every team sharing a repository with agents invents its own answer, and most answer only with more prose.

Two known uses The pattern is shown in two settings. The first is greenfield: one application, *taskboard*, a small Kanban service with a work-in-progress limit, built with the ladder from the start. We walk through its TypeScript implementation and show that Java and Python implementations with identical behaviour bind every rung equivalently (Griffin 2026). We call these implementations *receivers* of the pattern rather than its source, and work backwards from what each gains to the single capability they share. The second is a retrofit: *Tutors*, an open-source TypeScript learning platform in production, to which the rungs were added after the fact while coding agents were already contributing, and from which the release checks were then extracted into a reusable, product-independent harness (*Tutors contributors 2026a,b*). Together they show the ladder being established where none existed and being adopted where a codebase already had history.

Contributions This paper makes five contributions, each aimed at one or more of the gaps.

1. The Assertion Ladder, a pattern in the classic form (Alexander et al. 1977; Gamma et al. 1994) with six rungs and nine rung patterns, proposed as a shared vocabulary for how strongly a repository asserts each domain rule (G1, G2, G3; Section 3).
2. A unification of three kinds of intent, structure, behaviour and requirements, as three loads on the same ladder, with a closure check that ties requirements to scenarios and the observation that the pattern of an EARS requirement (Mavin et al. 2009) predicts the layer that should implement it (G4; Sections 3 and 4).
3. A greenfield receiver in TypeScript, with equivalent Java and Python receivers, showing which rungs each language

can reach and what it uses where it cannot (Section 4).

4. A retrofit of a production codebase, Tutors, and the extraction of its release checks into a reusable harness, showing how the ladder is adopted where it was not designed in (Section 5).
5. Violation seeding as a benchmark of a repository rather than of an agent, run before and after the behavioural and requirement rungs were added, and a pilot with eighteen coding agents (G5; Section 7).

2. Background

The Assertion Ladder draws on five bodies of work: the emergence of coding agents, architectural structure and its testing, behaviour and requirement specification, the psychology of learning, and lean thinking. This section introduces only what the later sections use.

2.1. Coding agents and the memoryless contributor

Code-generating models moved from completing a line (M. Chen et al. 2021) to resolving whole issues in real repositories (Jimenez et al. 2024). What made the second step possible was less the model than its harness: an agent that can list files, open them, edit them and run the tests behaves very differently from one that emits a single completion (Yang et al. 2024). Tool vendors now converge on a convention for telling such an agent how a repository works: a Markdown file at the root, loaded at the start of every session (AGENTS.md: A Simple, Open Format for Guiding Coding Agents n.d.). That file is the agent’s whole inheritance from previous sessions.

Three properties of this contributor matter for what follows. It is *capable*: given a clear pattern it will reproduce it across layers and languages. It is *memoryless*: nothing learned in one session survives to the next unless the repository records it. And it is *fast*: its output can exceed what a person can review line by line, which moves the person’s role from author to reviewer (Barke et al. 2023) and makes the quality of what passes review depend on what the reviewer can see.

2.2. Structure and its enforcement

Layering is one of the oldest structuring ideas in software (Dijkstra 1968); its rationale is information hiding, so that a decision likely to change is confined to one module (Parnas 1972). The layered architecture of enterprise applications, with a presentation layer, a service layer, a data-access layer and a domain model, is catalogued as a pattern (Buschmann et al. 1996; Fowler 2002), and domain-driven design places the domain model at the centre of attention (Evans 2003). Hexagonal and Clean Architecture invert the dependencies so that the domain depends on nothing and outer adapters depend on ports it owns (Cockburn 2005; Martin 2017).

Structure described only in prose decays. Architecture fitness functions make an architectural characteristic executable, so that a build can fail when it is violated (Ford et al. 2017); ArchUnit does this for Java dependencies by analysing bytecode in an

ordinary unit test (TNG Technology Consulting n.d.). Such rules are the structural core of our third rung.

2.3. Behaviour and requirements

Behaviour-driven development writes tests as examples of behaviour in the vocabulary of the domain, conventionally in the form *given* a context, *when* an event occurs, *then* an outcome is expected (North 2006); specification by example treats such examples as the living specification of the system (Adzic 2011).

The Easy Approach to Requirements Syntax (EARS) constrains natural-language requirements to a small set of templates, each distinguished by a keyword (Mavin et al. 2009): *ubiquitous* requirements (“The system shall. . .”), *state-driven* (“While. . .”), *event-driven* (“When. . .”), *optional feature* (“Where. . .”), *unwanted behaviour* (“If. . ., then. . .”) and *complex* combinations of these. EARS was designed to remove ambiguity for human readers. Section 3 argues that its keywords do more work in a repository shared with agents: they classify a rule in a way that predicts where its code belongs.

2.4. Learning: worked examples, scaffolding and cognitive load

Cognitive load theory distinguishes the load a task imposes because of its intrinsic difficulty from load imposed by how the task is presented, and holds that working memory is small (Sweller 1988). One of its best-established findings is the worked-example effect: novices learn a procedure faster by studying a solved example than by solving the problem themselves (Sweller & Cooper 1985). Scaffolding describes support that lets a learner do what they could not do alone, and that is withdrawn as competence grows (Wood et al. 1976).

A coding agent is not a novice in the usual sense; it is highly competent in general and entirely ignorant of this repository in particular. The analogy we use is limited to that second property. The repository must teach the particular, every session, within a context window that is large but not unlimited.

2.5. Lean thinking in software

Lean thinking began on the factory floor. The Toyota Production System, as described by its chief architect, set out to deliver what the customer wanted with as little waste as possible, and did so by building quality into each step of the work rather than inspecting it in at the end (Ohno 1988). Womack and Jones later restated it as five principles for any organisation: specify value from the customer’s point of view, map the value stream, make the work flow, let the customer pull it, and keep improving (Womack & Jones 1996). Software took up the same ideas early (Poppendieck & Poppendieck 2003, 2006), and they now sit beneath continuous delivery and DevOps (Forsgren et al. 2018). This paper uses four of them, value, standard work, value stream mapping and root cause analysis, and three supporting mechanisms.

Value and waste Lean starts from value: what the customer actually wants. Anything that does not serve it is waste. The Poppendiecks translated the seven wastes of manufacturing into software as partially done work, extra features, relearning,

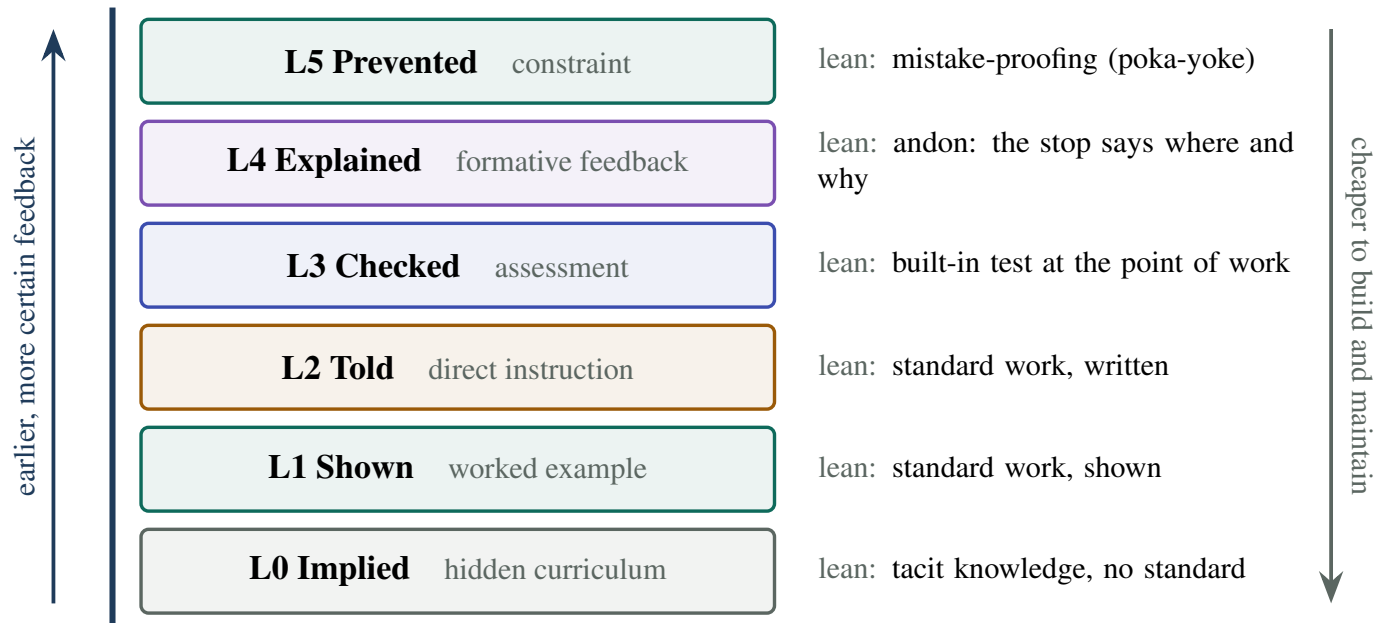


Figure 1 The Assertion Ladder. Each rung is a teaching move and a lean mechanism. Rules climb only as far as their risk warrants; as a rule reaches a checked rung, it is removed from the prose below it.

handoffs, task switching, delays and defects (Poppendieck & Poppendieck 2006). An observational study of software teams found an overlapping but different set, including rework, unnecessarily complex solutions, extraneous cognitive load and knowledge loss (Sedano et al. 2017). For a repository shared with agents we add one clarification. The customer is not the only party with an interest in a change; the domain has one too. A change that works for today’s user but spreads a rule across three places delivers value now and charges it back later as rework. In this paper, *value* is working behaviour that keeps the domain’s rules in one place and in its own words, and *waste* is anything that does not serve that end. Two wastes matter most. Relearning, or knowledge loss, is the memoryless agent’s normal condition: each session begins by rediscovering what the last one knew. Rework is what sprawl costs once it is finally noticed.

Standard work Standard work is the current best known way of doing a job, recorded so that everyone who does the job can follow it and anyone who finds a better way can change it. It is not a rulebook imposed from outside. In Toyota’s own account it is the foundation of improvement, because a job that nobody does the same way twice cannot be improved, only varied (Liker 2004). Spear and Bowen, studying Toyota plants, found the idea stated as a rule: all work is specified as to content, sequence, timing and outcome, and the specification carries built-in tests that signal a problem as soon as it occurs (Spear & Bowen 1999). In software, the nearest equivalents are a team’s agreed way of adding a feature, the examples it copies, and the tests that check the result.

Value stream mapping A value stream map draws every step a piece of work passes through from request to delivery, and marks which steps add value and which only wait, hand work

on or redo it (Rother & Shook 1999). The technique has been adapted to software, where the material is information and much of the waiting is invisible (Poppendieck & Poppendieck 2003; Khurum et al. 2014; Ali et al. 2016). The view we need is short. A change is written, verified, reviewed, merged and released. The later a defect is found along that stream, the more steps it has passed through and the more people have waited on it.

Root cause analysis and improvement Lean treats every problem as information about the process that produced it. Ohno’s practice was to ask *why* repeatedly, typically five times, until the answer is a cause that a change to the process can remove rather than a symptom to be patched (Ohno 1988). Toyota records such an analysis on a single sheet, the A3 report, which follows the plan–do–check–act cycle: state the problem, find its root cause, choose a countermeasure, and check whether it worked (Sobek II & Smalley 2008). The build–measure–learn loop of the lean startup applies the same cycle to product decisions (Ries 2011). Small improvements made this way, one after another, are what lean calls *kaizen*.

Supporting mechanisms Three further terms appear briefly. *Jidoka*, often translated as automation with a human touch, means that a machine stops itself when something goes wrong, so that nobody needs to watch it and no more bad work is made until the cause is fixed (Ohno 1988). In software, it is the failing build that blocks a merge. *Andon* is the signal that shows a stop has happened and where; any worker may also pull the andon cord to stop the line and ask for help. *Poka-yoke*, or mistake-proofing, designs a step so that a mistake cannot be made, or is caught at the moment it is made (Shingo 1986). In software, it is a type that will not compile when a case is missing.

Lean in knowledge work, and its limits Lean transfers to knowledge work, but not unchanged. At a large software services firm, projects run on lean principles outperformed comparable projects on most measures, yet uncertain tasks, invisible work in progress and design problems that surface only during implementation all made the manufacturing tools harder to apply (Staats et al. 2011). A review of practitioner reports found lean applied within agile teams in many ways and for different purposes (Wang et al. 2012). Reinertsen argues more sharply that product development is not repetitive manufacturing: some of its variability creates value, and standardising that variability destroys the value (Reinertsen 2009). We take the point. The ladder standardises only the part of software work that ought to repeat, such as where a rule lives, how an error is translated and how a scenario cites a requirement, and leaves design to the people and agents doing it.

2.6. Patterns

A pattern names a recurring problem in a context, the forces that make it hard, and a solution that balances them (Alexander et al. 1977). The form was brought to software design by the Gang of Four (Gamma et al. 1994) and to architecture by the POSA series (Buschmann et al. 1996). We use it for the Assertion Ladder because the ladder is not an algorithm or a tool but a way of resolving forces that every team sharing a repository with agents will meet.

3. The Assertion Ladder

This section states the pattern, describes each rung, names the smaller patterns that realise the rungs, and shows that structure, behaviour and requirements are three loads carried by the same ladder rather than three separate techniques.

3.1. The pattern

Name	Assertion Ladder (also: Graduated Assertion).
Intent	Keep a technical domain intact when coding agents contribute, by asserting each domain rule at the weakest level that still covers its risk.
Context	A repository shared by people and coding agents that start every session with no memory. Changes arrive faster than anyone can review them closely.
Problem	How do a domain's rules survive contributors who cannot remember them and reviewers who cannot read everything?
Forces	See Figure 2. – A rule nobody wrote down is lost at the next session. – Written guidance grows until nobody reads it, and competes with everything else in an agent's context. – Every executable check costs effort to write, time in the build, and maintenance when the domain changes.

- The strongest assertions, those made by the type system, depend on the language.
- Too much constraint makes the repository hard to change. Ease of use is a force, not an afterthought.
- A check that an agent can edit is a check that an agent can weaken.

Solution

For each domain rule, choose a rung. Climb until the rule's risk is covered, then stop. When a rule reaches a checked rung, remove it from the prose so that the declaration stays short. Run every check behind one command, and put the checks themselves behind a human boundary.

Consequences

Feedback arrives early and names the fix; the manifest stays short; intent can be reviewed separately from code. In exchange there are more tests to maintain, the top rung differs between languages, and a rule that climbs further than its risk warrants makes change expensive.

3.2. The rungs

L0 Implied The rule exists only in someone's head or in habit. The agent experiences nothing, and guesses. The guess is usually plausible, which is what makes L0 dangerous: nothing about the result looks wrong. In lean terms this is tacit knowledge with no standard: there is nothing to follow and nothing to improve, and every session pays the waste of relearning. In teaching terms it is the hidden curriculum.

L1 Shown One complete example embodies the rule: a vertical slice from the domain to the endpoint, small enough to read whole. Agents, like novices, copy structure before they understand it (Sweller & Cooper 1985), so the slice is the cheapest effective assertion there is. In lean terms the slice is standard work shown rather than written: the team's current best way to add a feature, in a form a contributor can copy. Its weakness is that the learner may copy the wrong detail, or generalise from an accident of the example.

L2 Told A short declaration states the rule, in a file the agent's tooling loads at the start of every session (*AGENTS.md: A Simple, Open Format for Guiding Coding Agents* n.d.). Prose can say what code cannot: what needs a person's decision, which of two acceptable choices the team prefers, why a rule exists. In lean terms it is the written half of standard work. Its weakness is that it can be forgotten, misread, or crowded out; and it grows.

L3 Checked An executable rule fails the build when the rule is broken. This is the rung of architecture tests, contract tests and behavioural scenarios. The agent experiences a red suite and must work out why. In lean terms it is the built-in test that Spear and Bowen found attached to Toyota's standard work (Spear & Bowen 1999): the defect is found where it is made, by the contributor who made it, before the change moves on to a

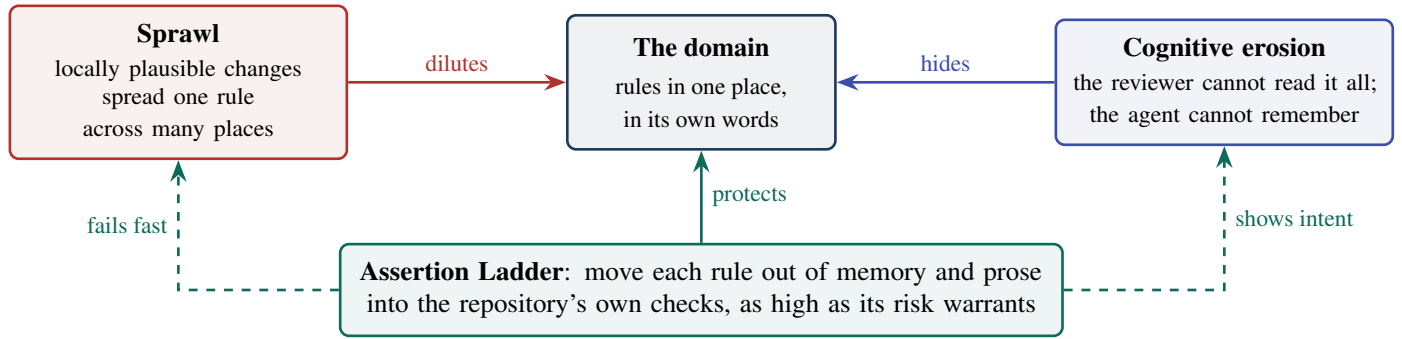


Figure 2 Two forces act on a domain once agents contribute. Sprawl spreads a rule across places; cognitive erosion hides the spread from both reviewer and agent. The ladder answers both by moving rules out of memory and prose into the repository’s own checks.

reviewer, rather than by a separate inspection further down the stream.

L4 Explained The check’s failure names the rule, the offender and the fix. A bare assertion failure tells the agent *that* it is wrong; an explained failure tells it *what to do instead*, which is the difference between summative and formative feedback. In lean terms a bare failure is a stop; an explained failure is a stop that, like an andon signal, shows where the problem is and why, and names the countermeasure. It costs a sentence per rule.

L5 Prevented The wrong form does not compile. A sealed hierarchy with an exhaustive switch, or a map keyed by a closed union type, makes an unmapped error impossible to write. This is mistake-proofing (poka-yoke) of the preventive kind: the fixture admits the part only one way, so the mistake is never made, where a check at L3 catches it once it has been. It is also the rung that depends most on the language.

3.3. Nine rung patterns

Each rung is realised by one or more smaller patterns, listed in Table 1. Two are not tied to a rung: SINGLE GATE, which runs every check behind one command so that “finished” has one meaning, and HUMAN BOUNDARY, which lists the few changes an agent may not make alone, including changes to the checks themselves.

CLOSURE CHECK deserves a note because it recurs more than any other. An *every X has a Y* assertion keeps two lists in step without either list knowing about the other: every domain error has an HTTP status; every repository implementation passes the same contract; every requirement has a scenario and every scenario cites a requirement. It is the cheapest way to make a growing set safe to extend, which is exactly what an agent does.

3.4. Three kinds of intent, one ladder

Architecture tests, behaviour-driven development and structured requirements are usually presented as separate practices with separate literatures. In a repository shared with agents they turn out to be the same move applied to three kinds of intent (Table 2). *Structure* says where code may live and what it may touch. *Behaviour* says what the system does, in the domain’s

own words. *Requirements* say why, once, in a form that can be traced. Each is declared, exemplified and enforced, and each has its own closure check.

3.5. EARS as a placement oracle

The most useful property of EARS in this setting was not one it was designed for. Its keywords classify a rule, and the class predicts where the rule’s code belongs (Table 3). An *unwanted behaviour* about one entity (“If a requested move is not allowed from the task’s status, then. . .”) is a guard on that entity that raises a domain error. A *state-driven* rule that spans many entities (“While the work-in-progress limit is reached. . .”) needs a collaborator and belongs in a service. An *event-driven* rule whose response is to tell someone (“When a task moves to done, the board shall announce. . .”) is a domain event published by the service with the effect in a handler. An *optional feature* (“Where a limit is configured. . .”) is a field on the one settings object. A *ubiquitous* rule (“The board shall answer every refused request with a problem document”) is a cross-cutting invariant: one place, guarded by a check.

The mapping matters because the commonest placement mistake an agent makes is to put a correct rule in the wrong place, and a behavioural test cannot see that mistake (Section 7). A requirement written in EARS carries its placement with it. We present the mapping as a hypothesis supported by the ten taskboard rules and by one pilot run, not as a law.

3.6. Compression and fading

The solution says to climb only as far as a rule’s risk warrants, and to remove a rule from the prose once a check covers it. Both clauses are about ease of use, the force most easily forgotten. A manifest that repeats what the tests enforce is waste in the lean sense: it costs reading time every session and adds nothing the gate does not already guarantee. Standard work is meant to change as the team learns, and here it changes by subtraction as well as addition: once a rule is built into a check, the written standard no longer needs to carry it. Instructional design calls the same move *fading*: scaffolding is withdrawn as the learner no longer needs it (Wood et al. 1976). In the receivers the manifest says so of itself: “anything a test can check lives in the tests, not here.” Section 7 reports what the added rungs cost in

Table 1 The rung patterns. Each answers one recurring problem.

Rung	Pattern	Problem	Solution
L1	WORKED SLICE	How does a newcomer learn the shape of a feature?	One complete vertical slice, small enough to read whole.
L2	SHORT MANIFEST	How is what cannot be tested passed on?	One file of a few hundred words, loaded every session.
L3	EXECUTABLE RULE	How is a structural rule kept when nobody remembers it?	A test that fails when the rule is broken.
L3	CLOSURE CHECK	How do two lists stay in step?	Assert that every X has a Y.
L3	HONEST DOUBLE	How do fast tests stay truthful?	One contract, run against the real implementation and its stand-in.
L4	TEACHING FAILURE	How does a red build lead to the right fix?	The failure names the rule, the offender and the remedy.
L5	UNREPRESENTABLE MISTAKE	How is a mistake stopped before any test runs?	Types that make the wrong form fail to compile.
all	SINGLE GATE	When is work finished?	One command runs every rung; green means done.
all	HUMAN BOUNDARY	Who may change the ladder itself?	A short list of changes that need a person first.

Table 2 The same three carriers hold three kinds of intent. Text colour follows the intent throughout the paper.

Intent	Declared (L2)	Exemplified (L1)	Enforced (L3–L5)
Structure: where code may live	a placement table in the manifest	the move-task slice, domain to route	architecture rules; exhaustive error mapping
Behaviour: what the system does	one convention line: scenarios cite ids and read given / when / then	the work-in-progress scenario	the scenarios, and the repository contract
Requirements: why	REQUIREMENTS.md, one EARS sentence per rule	ten taskboard rules with ids	the requirement closure check

words.

4. A greenfield receiver in TypeScript

A pattern is only as credible as its known uses. This section presents the first: one application, *taskboard*, built with the ladder from the start. It exists three times with identical behaviour, in TypeScript, Java and Python. We call each a *receiver* because it does not define the Assertion Ladder; it takes the ladder on in the idiom of its language and gains something from it. We walk through the TypeScript receiver in detail, since it shares a language with the production retrofit in Section 5, and then show that the Java and Python receivers are equivalent: every rung pattern is bound in each, sometimes on a different rung. The second known use, a retrofit of a production codebase, follows in Section 5.

4.1. The application

Taskboard is a Kanban API with one entity. A task moves from *todo* to *in progress* to *done*, and may move back from *in progress* to *todo*. No more than a configured number of tasks may be in progress at once. When a task is completed the team

is told. Every refused request is answered with an RFC 9457 problem document carrying a stable error code. The application is deliberately small: 283 lines of application code in Java, 223 in Python and 242 in TypeScript, down from 637, 729 and 957 in a first iteration that used dependency-injection containers, object-relational mappers and a second entity (Griffin et al. 2026). Small enough to read whole is a property the ladder depends on: the WORKED SLICE only works if the slice can be read in one sitting.

Each receiver is organised in the same six layers: *api*, *services*, *repositories*, *domain*, and the side layers *events* and *config*, with one composition root. Imports point only downwards, the domain imports nothing but the language’s standard library, and each outside capability (HTTP, SQL, the environment, logging) belongs to exactly one layer.

4.2. A steel thread through all three kinds of intent

Figure 3 follows one rule, the work-in-progress limit, from its requirement to the gate. The requirement is written once in EARS:

R-WIP-1 (state-driven). While WIP_LIMIT tasks are in progress, when another task is moved to *in_progress*, the

Table 3 From EARS pattern to layer, with the taskboard rule that exercises each row.

EARS pattern	Implement as	Taskboard rule
Unwanted, one entity	a guard on the entity raising a domain error	R-FLOW-2: an invalid move is refused
State-driven, many entities	a service method with a collaborator	R-WIP-1: the work-in-progress limit
Event-driven, tells someone	a domain event; the effect in a handler	R-DONE-1: completion is announced
Optional feature	a field on the settings object	the configurable limit
Ubiquitous	one place, guarded by a closure or architecture check	R-ERR-1: every refusal is a problem document

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1   board shall refuse with wip_limit_exceeded and leave
2   the task where it was.

3   Its pattern predicts a service method, because the rule needs to
4   count other tasks. The scenario that covers it cites its id and
5   reads as given / when / then (Listing 1). The domain error it
6   raises carries a code and no HTTP; one handler maps it to a
7   status, and a closure check refuses a new error without one. The
8   limit arrives through a constructor from the one settings object.
9   Every step is checked by the same command.

10  test("R-WIP-1: given a full board, when another
11      task is
12      started, then it is refused and stays in
13      todo", async () => {
14      const first = await service.createTask("First")
15      ;
16      const second = await service.createTask("Second")
17      );
18      await service.moveTask(first.id, "in_progress")
19      ;
20      await expect(service.moveTask(second.id, "in_progress"))
21      .rejects.toThrow(WipLimitExceeded);
22      expect((await service.getTask(second.id)).status).toBe("todo");
23  });

```

Listing 1 A scenario in the TypeScript receiver. The title cites the requirement and reads given / when / then; the requirement check fails if either is missing.

4.3. The TypeScript binding path

Read rung by rung, the TypeScript receiver binds the ladder as follows. **L1:** the WORKED SLICE is moveTask, from the service down to the route. **L2:** the SHORT MANIFEST is an AGENTS.md of 468 words, and the rules are ten EARS requirements in REQUIREMENTS.md. **L3:** five EXECUTABLE RULES over source text, a requirement CLOSURE CHECK, and a repository contract run against SQLite and an in-memory HONEST DOUBLE, all behind one SINGLE GATE, npm test. **L4:** every assertion carries a TEACHING FAILURE message that names the rule and the fix (Listing 2). **L5:** the status map is a Record keyed by a closed union of error codes, so a missing entry is a compile error, an UNREPRESENTABLE MISTAKE (Listing 3). The HUMAN BOUNDARY lists three changes that need a person first: dependencies, the rules and manifest, and schema or endpoint shape.

```

1  // Ambient capabilities, and the one layer that
2  // may touch each.
3  const RESERVED: [RegExp, string][] = [
4  [/process\.env/, "config"],
5  [/from "node:sqlite"/, "repositories"],
6  [/from "fastify"/, "api"],
7  [/console\.\/, "events"],
8  ];
9  expect(misplaced, "move that work into the owning
10 layer; " +
11 "reach it through a constructor argument or a
12 domain event")
13 .toEqual([]);

14 // Routes translate between HTTP and a service
15 // call.
16 const DECISION = /\b(if|for|while|switch|try|
17 throw)\b\.(filter|find|some|every|reduce)
18 \(\);
19 expect(decision, "routes translate; put the rule
20 on the " +
21 "entity if it concerns one task, otherwise in
22 the service")
23 .toBeUndefined();

```

Listing 2 Two executable rules with teaching failures in the TypeScript receiver. The second closes a gap that no behavioural test can see (Section 7).

```

1  // Record<ErrorCode, number> is exhaustive: a new
2  // error code
3  // will not compile until it has a status.
4  const STATUS_BY_CODE: Record<ErrorCode, number> =
5  {
6  task_not_found: 404,
7  invalid_transition: 409,
8  wip_limit_exceeded: 409,
9  };

```

Listing 3 Unrepresentable Mistake in the TypeScript receiver.

Its architecture rules are the cheapest of the three receivers' styles, regular expressions over source text, and the least precise. They are adequate for a codebase this small and would need a parser for a larger one; Section 5 shows what a production TypeScript codebase used instead.

4.4. Equivalence in Java and Python

Table 4 shows that the Java and Python receivers bind every rung pattern the TypeScript receiver binds. Reading a row shows one pattern bound three ways; reading a column shows one receiver's complete path. The behaviour is identical: the same ten requirements, the same scenarios, and the same fourteen

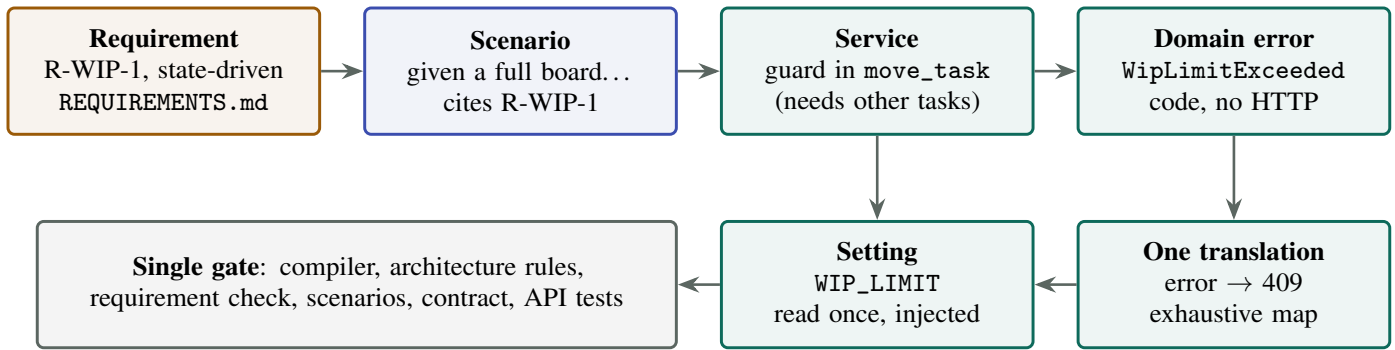


Figure 3 The work-in-progress limit as a steel thread. Colours mark the kind of intent each step carries; every step is checked by the single gate.

1 seeded mistakes caught (Section 7). What differs is the rung on
2 which a language can carry a rule.

3 **Java** The Java receiver reaches the top of the ladder most
4 often. Its domain errors form a sealed hierarchy, and the han-
5 dler that maps them to HTTP uses a `switch` with no default
6 branch, so the TypeScript Record has an exact equivalent at
7 L5. Its architecture rules are ArchUnit rules over bytecode, the
8 most precise of the three styles, and each carries a because
9 clause that ArchUnit prints on failure; its rule for side effects,
10 the counterpart of the TypeScript `console` entry, reads “side
11 effects (logs, mail, audit) listen for a domain event; publish one
12 instead”. Its requirement check imports the test classes them-
13 selves and asserts that every scenario’s `@DisplayName` opens
14 with a requirement id, where TypeScript reads test titles.

15 **Python** Python cannot make an unmapped error unrepre-
16 sentable, so the Python receiver binds that rule one rung lower: a
17 `CLOSURE CHECK` asserts that every subclass of `DomainError`
18 has an entry in the status map. It compensates elsewhere. Its
19 architecture rules walk the syntax tree, a stricter version of
20 the TypeScript text rules: the rule that a route decides nothing
21 is checked against `If`, `loop`, `comprehension`, `Raise` and `Try`
22 nodes rather than keywords. The strict type checker runs inside
23 the same gate. Scenarios cite requirements in docstrings where
24 TypeScript uses titles.

25 4.5. Layered, not Clean, and what that costs

26 All three receivers are layered rather than hexagonal or Clean.
27 The persistence interface lives beside its implementation in the
28 repositories layer rather than as a port owned by the core, so
29 dependency inversion does not stop a service from reaching
30 SQL. Capability confinement does: a rule that only the reposi-
31 tories layer may use the database driver, only the api layer the
32 web framework, only the config layer the environment, and
33 only the events layer logging. We read this as a finding about
34 the ladder rather than about architecture styles. A handful of
35 `EXECUTABLE RULES` recovers most of the domain protection
36 Clean Architecture provides, without the ports, adapters and in-
37 teractors an agent would otherwise have to learn and reproduce
38 in every feature. The trade is ceremony against rules; the ladder
39 lets a team make it explicitly.

40 4.6. What each receiver gains

41 Working backwards from the receivers, each gains the same
42 four things. The domain stays in one place, which the seeding
43 results in Section 7 measure directly. A new feature has a recipe:
44 state the rule in EARS, let its pattern choose the layer, copy the
45 slice, run the gate. Failures say what to do. And the manifest a
46 new contributor reads is under 500 words, because everything
47 checkable lives in the checks. The receivers differ only in which
48 rung carries which rule, and those differences are themselves
49 instructive: the same rule set, bound by different languages,
50 lands on different rungs.

51 5. Retrofitting a production codebase: Tutors

52 The receivers of Section 4 were built with the ladder in mind.
53 This section asks whether the ladder can be climbed by a code-
54 base that was not: an open-source teaching platform in pro-
55 duction, with years of history and few tests, into which coding
56 agents were already contributing. It reports what was retrofitted,
57 in what order, what it reached and what it could not, and how its
58 release gate was extracted into a separate, reusable repository.
59 Every figure below is read from the two public repositories, the
60 monorepo at commit 865d02f ([Tutors contributors 2026a](#)) and
61 the release harness at version 1.15.0, commit c324525 ([Tutors contributors 2026b](#)). Where a figure is our own count rather
62 than one the repositories record, we say so.
63

64 5.1. Context

65 Tutors is a course reader and learning platform. Lectur-
66 ers write courses in Markdown; a generator turns a course
67 folder into one `tutors.json`; four SvelteKit applications
68 (reader, catalogue, live and time) render it, track pres-
69 ence and report learning time. The stack is TypeScript, Svelte 5
70 and SvelteKit, six Deno packages published to JSR, pnpm
71 workspaces and Supabase (`README.md`, `ARCHITECTURE.md`).
72 The monorepo holds the four applications and 24 packages.
73 By our count, product code in `apps/` and `packages/` is about
74 24,500 lines of TypeScript and Svelte, excluding tests, fixtures,
75 declaration files and a vendored PDF worker; test code is about
76 33,500 lines.

77 The monorepo was founded on 27 July 2026 by consolidating
78 several repositories; the original reader repository has history

Table 4 Binding paths of the three receivers. TypeScript is the worked example; Java and Python show equivalence.

Rung pattern	TypeScript (Fastify)	Java (Spring Boot)	Python (FastAPI)
WORKED SLICE	moveTask, service to route	moveTask, service to controller	move_task, service to route
SHORT MANIFEST	AGENTS.md, 468 words	AGENTS.md, 480 words	AGENTS.md, 427 words
EXECUTABLE RULE	source text, 5 rules	ArchUnit over bytecode, 8 rules	AST walk, 7 rules incl. strict type check
CLOSURE CHECK	errors by the compiler; requirements by a test	errors by the compiler; requirements by RequirementsTest	errors and requirements by tests
HONEST DOUBLE	contract run on SQLite and in-memory	contract run on JDBC and in-memory	contract run on SQLite and in-memory
TEACHING FAILURE	a message on every assertion	because(...) on every rule	a message on every assertion
UNREPRESENTABLE MISTAKE	Record<ErrorCode, number>	sealed DomainException, exhaustive switch	not available; bound at L3
SINGLE GATE	npm test	mvn verify	pytest
HUMAN BOUNDARY	the same three items in each manifest: dependencies; the rules and manifest; schema and endpoint shape		

from 2022 (guides/evolution/tutors-revolution.md). On 1 June 2026 it had no tests, no CI workflow and no Dockerfile, and the whole ecosystem had 27 tests (guides/testing/journey.md). Coding agents took part in the retrofit throughout: by our count of git trailers, 329 of the monorepo's 483 non-merge commits, and all 144 of the harness's, were authored or co-authored by an agent. The retrofit is therefore a second known use of the paper's setting, not a laboratory one: a domain protected while agents change it.

5.2. The retrofit, rung by rung

Table 5 maps each rung pattern to the artefact that binds it in Tutors. The order in which they arrived is itself informative. The first moves, in early August, were to replace call-sequence mocks with real implementations and a data-driven stand-in carrying 27 validation tests of its own (commit 786f1fe), and to clear 138 type errors (95064ea). These are the cheapest rungs: they make existing tests honest before new ones are added. Structure came next. On 16 September a "testing runway" added architecture rules, suite-health lints and completeness checks as tiers A to O, and one aggregate CI job, ci-success, as the required status (c57b9eb, 3bf806b). Requirements came last. Feature files had existed since August, but on 17 September all 24 were found to be documentation only; binding them to product code (adffc55) found bugs in search, sort and logging (guides/evolution/tutors-revolution.md, "Spec-first"). EARS Rule: blocks with permanent ids and an audit followed on 21 September, and on 26 September coverage and mutation floors.

Structure The architecture rules state Tutors' layering (foundation, core, feature, three UI layers, apps) as one list in .dependency-cruiser.cjs, from which one rule per layer is generated. Each rule carries a comment that says why and what to do instead, and each has a negative fixture under tests/architecture/fixtures/, a small workspace

that breaks it on purpose. The test asserts that every fixture is caught, that no clean file is flagged, and that the live repository adds no violation beyond a baseline (tests/architecture/dependency-rules.test.ts). Tutors is layered, like the receivers, and enforces the layering by the same move: an EXECUTABLE RULE rather than ports and adapters.

Behaviour and requirements Behaviour and requirements are carried together. A requirement is the title of a Gherkin Rule:, written in EARS (Mavin et al. 2009) with exactly one "shall", a named system and a permanent id (@rule-0110); its scenarios are the proof, bound to product code by vitest-cucumber. Listing 4 shows one. The audit is a CLOSURE CHECK in both directions, and names each failure with a code and its fix (.claude/skills/ears-gherkin-dev/references/audit-codes.md). It reaches past the repository: a release claim cites a Rule id, and npnm check:release-claims fails when no feature defines it (guides/testing/EARS-METHODOLOGY.md). A requirement is thus traced from its sentence to its scenario, its changelog line and the observable difference it licenses. Unlike the receivers, Tutors files Rules by persona rather than by layer, so it does not use the EARS pattern as a placement oracle (Table 3).

```
@rule-0111 @ears-unwanted
Rule: If total or per-package test coverage falls
    below
    its recorded floor, then tutors shall fail the
    coverage
    check and name the scope and metric.
Scenario: A package below its floor fails the
    check
    Given recorded coverage floors where
        "packages/jsr/model/src/**" holds
        statements at 83 percent
    When the measured statement coverage of
        "packages/jsr/model/src" is 82.9 percent
```

Table 5 The rung patterns as bound in Tutors, with the reach of each. Paths are in the monorepo unless marked *harness*.

Rung pattern	Tutors artefact	Reach
WORKED SLICE (L1)	docs/COURSE-PAGE-WALKTHROUGH.md: one URL followed through every file to the rendered cards	a slice in prose, not a small runnable exemplar
SHORT MANIFEST (L2)	CONTRIBUTING.md (2,043 words), ARCHITECTURE.md; the ears-gherkin-dev agent skill (962 words)	no AGENTS.md or CLAUDE.md; the skill loads on demand, not every session
EXECUTABLE RULE (L3)	.dependency-cruiser.cjs: ten rules (six layer rules, no app-to-app, no relative cross-workspace import, primitives off data packages, no cross-package cycle); knip; manifest parity	new code at L3; three older violations held in known-violations.txt
CLOSURE CHECK (L3)	EARS audit, scripts/checks/ears-audit.ts: every Rule has a scenario, no scenario outside a Rule, every Rule bound, every UI scenario has a Playwright test and the reverse; every claim cites a Rule that exists	95 Rules, 193 scenarios in 34 feature files (our count); audit required in ci-success since 23 Sep
HONEST DOUBLE (L3)	MockSupabaseClient with its own tests; <i>harness</i> : a second, in-memory backend behind each seam, driving the same collector	data-driven, not one contract run on both sides
TEACHING FAILURE (L4)	a comment on every dependency rule; ratchet messages (“Fix them rather than adding them to ...”); an audit code table with the fix per code	L4 wherever a check exists
UNREPRESENTABLE MISTAKE (L5)	strict TypeScript and blocking svelte-check; Zod validation of the environment	weak: loTypes is typed string[], so LoType is string
SINGLE GATE	locally pnpm lint, pnpm test, pnpm check; in CI one required job over eleven; at release, the harness Gate	three commands, then one status
HUMAN BOUNDARY	CODEOWNERS on release/claims.yaml and release/SOP.md; <i>harness</i> : CODEOWNERS on masks, mutants, gate.ts and the contract; a broad claim needs a named person	declared; review not yet practised

```

110 Then the coverage check shall report "below-
111 floor:
112   packages/jsr/model/src/** statements 82.9%
113   < floor 83%"

```

Listing 4 An EARS Rule in Tutors, from the test-quality ratchets feature file. The audit checks its form and tag; its scenarios check its truth. Steps are wrapped for width.

held 169 entries on 21 September and 167 on 28 September (journey.md). Coverage and mutation floors ratchet the other way: each is the measured value rounded down, and a floor two or more points below the measurement fails as stale (tests/suite-health/coverage-floors.json, tests/mutation/mutation-floors.json). A ratchet lets a rule sit at L3 for new code on the day it arrives while old code is paid down from a ledger. The receivers never needed this, and a codebase with history cannot do without it.

The second addition answers the paper’s own evaluation method. Section 7 seeded mistakes from outside the receivers to show the checks can fail. Tutors keeps that evidence inside the repository. “No tier without a negative fixture” is a stated rule (journey.md). Mutation testing covers twelve targeted modules, breaking below 90% with a measured score of 93.6%, and a nightly run over every library module, whose first baseline scored 58.5% over 95 modules (guides/testing/MUTATION-TESTING.md). The harness applies the same rule to itself: a new engine does not merge without an A/A test, a planted change it catches and a change it must not flag (*harness* docs/user-guide/09-extending.md).

5.4. Abstraction: the release harness

The last rung Tutors added sits outside the monorepo (Figure 4). Its precursor was an in-repository check that the generator’s out-

1 put matched production's, byte for byte (`tests/release/`
2 `RELEASE-TESTING.md`). The release harness generalises
3 that check to everything a user can observe. It runs a candidate's
4 images beside production's, drives the same scripted journeys
5 through both, captures nineteen kinds of artefact (DOM, screen-
6 shots, network, console, headers, accessibility, focus order, tim-
7 ing, metrics, logs, persisted writes, SBOM, vulnerabilities, con-
8 tainer posture and others), masks known noise, and fails unless
9 every difference is covered by a claim (*harness* README.md).
10 A claim names an artefact, a scope and the Rule or changelog
11 entry that justifies it (`release/claims.yaml`). This is a CLO-
12 SURE CHECK over the release: every observable difference has
13 a claim, and a claim that matches nothing is reported as stale.

14 Why a separate repository? The reason given is ours in
15 Section 3: a check that an agent can edit is a check that an
16 agent can weaken. The harness "consumes the monorepo's
17 published images ... and knows nothing about their source, so
18 it can compare any two tags ... and cannot be quietly weak-
19 ened by the PR it is judging" (*harness* README.md; decision
20 1 in `docs/architecture/06-decisions.md`). The re-
21 ceivers bind the HUMAN BOUNDARY at L2, as a line in the
22 manifest. Extraction binds part of it structurally: the gate's
23 code, its masks and its mutants live where the change under
24 review cannot reach them, behind their own CODEOWNERS.
25 The harness also holds itself to the discipline it imposes. It may
26 fail a release only after a clean production-against-production
27 run less than seven days old, and it must catch ten planted re-
28 gressions, each attributed to the expected artefact, weekly and
29 on any change to an engine (*harness* `mutants/mutants.yaml`,
30 `src/gate.ts`).

31 What is generic is the contract, not the stack. The integration
32 contract is versioned and "is what `tutors-sdk/tutors`
33 `-mono-repo` (or anything else) may build against"; a test fails
34 when document and code drift (*harness* `docs/contract.md`).
35 An adopter supplies signed images, a claims file and optionally
36 a `rules.json` of Rule ids, and triggers the harness by dispatch
37 (*harness* `docs/monorepo/README.md`). The four applica-
38 tion names, the journeys and the fixture stubs are still Tutors'.
39 We therefore read the harness as the paradigm made reusable at
40 the level of its contract, and not yet as a tool demonstrated on a
41 second product.

42 **Lean, named in the code** The harness is explicit about its lean
43 vocabulary, and each term points to a mechanism, as our plan
44 requires (*harness* `docs/lean.md`; `release/SOP.md`). *Jidoka*:
45 the Gate stops the line, and an advisory confidence score may
46 never override it ("a FAIL is a FAIL at RCS 99"). *Standard*
47 *work*: a twelve-step release SOP with an owner, an input and a
48 done-when per step, in which steps 2 to 7 run as one command
49 that stops before step 8, the one step that must stay human.
50 *Gemba*: a reviewer's glance of at most seven ranked places, each
51 marked verified, disputed or escalated. *Kaizen* and root cause
52 analysis: *harness why* opens a 5 Whys whose first answer is
53 the harness's own trace, rejects "human error" as an answer,
54 and requires one of seven countermeasure kinds, a new mutant
55 being preferred. *Value stream*: *harness a3* draws an A3 whose
56 current condition is a value stream map of a change from merge

to production. Compression is written in as well: "six journeys
in three sets, not a hundred; ten mutants, not thirty. ... Add
a journey only when a real regression escaped that a journey
would have caught" (*harness* README.md, "Where to stop").

5.5. What the retrofit shows

Three results are measured by the repositories themselves. First,
the ladder can be climbed on a production codebase in weeks:
from 27 tests on 1 June to 2,140 declared tests, 184 bound
scenarios under 89 EARS Rules and 51 CI jobs on 28 September
(*journey.md*). Second, the release rung finds what the lower
rungs miss. The harness found four production bugs that no test
had, including an authentication page that answered 500 and a
first visit that was never recorded (*journey.md*, "Judge every
release against production"). Third, it applies to a real change
the same root-cause discipline Section 8 applies to the seeding
misses: three 5 Whys are open in the register, each ending in a
countermeasure to the system rather than to a person (*harness*
kaizen/README.md).

Two results are demonstrated rather than measured. The
three kinds of intent survive translation to a codebase the ladder
did not shape, and a requirement id can carry intent from a sen-
tence through a scenario to a release decision. And the retrofit
adds two things the receivers lack and a greenfield project would
benefit from: ratchets for legacy debt, and self-tests that keep
the seeding evidence inside the repository.

We narrow the example to TypeScript because Tutors is
written in it. The mechanisms have counterparts in the other
receivers' stacks: the dependency rules correspond to ArchUnit
and to the Python receiver's syntax-tree rules, and mutation tools
exist for the JVM and for Python. The harness judges images
and claims, not source, so we expect a Java or Python service to
adopt it through the same contract; that is a hypothesis, not yet
tried.

5.6. Limits

The retrofit does not reach every rung, and its own documents
say so. There is no short manifest loaded every session; the
nearest are a 2,043-word contributing guide and one agent
skill loaded on demand. L5 is weak where the domain is rich-
est: the learning-object type that the whole model turns on re-
solves to `string` (`packages/jsr/model/src/types/`
`type-utils.ts`). The global coverage floor was 58% of
statements against a 90% target on 28 September, and tier F,
authorisation, is planned but not built (*journey.md*). The HU-
MAN BOUNDARY is declared but not yet practised: none of
the 55 pull requests merged since release 16.2.2 had an ap-
proving review, and "the CI gate, the ratchets and the harness
stand in for review; they do not replace it" (*journey.md*). The
release-confidence tooling is built, but its evidence gates need
real releases: the scoreboard is empty (*harness* `scoreboard/`
`releases.jsonl`), no 5 Whys is yet closed, and the daily
forecast of main against production was FAIL with 892 un-
claimed differences on 28 September, most of them an in-
tended image change that nobody had claimed (`guides/`
`evolution/tutors-revolution.md`). Finally, Tutors
is one codebase, the retrofit was carried out largely by two of

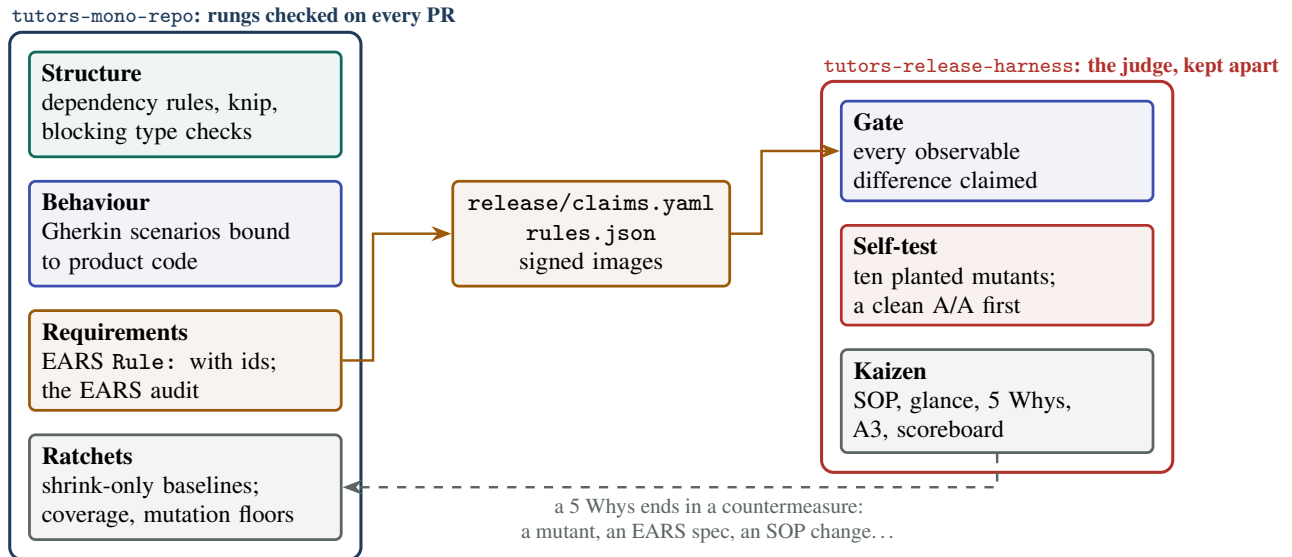


Figure 4 Tutors keeps structure, behaviour and requirement checks in the monorepo and ratchets their debt. The judge of a release is a separate repository that reads only signed images, the claims and the Rule ids, and whose 5 Whys feed countermeasures back into both.

1 this paper’s authors working with agents, and it was not de-
2 signed as an experiment. We report it as a known use of the
3 pattern, not as a test of it.

6. Use cases

5 The receivers show the pattern bound in code. This section
6 states, independently of any language, the situations in which
7 a domain is at risk once agents contribute, so that the pattern
8 can be applied to domains other than task boards. Each use
9 case names a situation, the threat, the invariant at stake and
10 the teaching move that answers it. Table 6 lists twelve in six
11 families and Figure 5 places each on the ladder, both in the
12 receivers as first published and as they stand after the work
13 described here.

6.1. Placement

15 Placement failures are the signature of sprawl. In UC1 an agent
16 with no prior session implements a feature end to end; the risk is
17 not that the feature fails but that each layer gets a plausible and
18 inconsistent piece of it. The answer is a worked slice to copy,
19 followed by architecture rules to pass. UC2 and UC3 are subtler
20 because the mistake preserves behaviour. A notification called
21 inline in a service, instead of published as an event, produces
22 the same log line; a work-in-progress check copied into a route
23 produces the same refusal. No behavioural test can tell the
24 difference, and at commit 8c4a74a nothing in any receiver did.
25 They were closed by two EXECUTABLE RULES, that only the
26 events layer may log or notify and that routes make no decisions,
27 each with a TEACHING FAILURE naming where the code should
28 go.

6.2. Contamination

30 In UC4 a framework annotation looks convenient on a domain
31 object; in UC5 a service needs a setting or a quick query. Both

are answered by capability confinement (Section 4), and both
were already checked before this work; they gained explained
failures.

6.3. Contract integrity

UC6, a new failure mode, is the one use case the receivers
answer at L5 in two of three languages: the new error does not
compile until it has a status. UC7 is the test double that lies,
answered by an HONEST DOUBLE. It has a history: in the first
seeding run the contract created tasks titled First, Second and
Third, whose alphabetical order equals their creation order, so
a double that sorted by title could not be caught. Changing the
titles to Write, Review and Publish fixed it. The episode is a
small instance of kaizen, and a reminder that a check is only as
good as its fixture.

6.4. Harness erosion

UC8 and UC9 concern the ladder itself. An agent whose build
is red has an easy way to make it green: relax the rule. No agent
in our pilot did so, but only the manifest forbade it (L2). Raising
UC8 to L3 is a matter of repository settings rather than tests, for
example by requiring a named person’s review for any change
to the rule files. Dependencies (UC9) are similar, and can be
checked by pinning the dependency list in a test.

6.5. Cognitive load

UC10 is the reviewer’s problem. When a change adds a feature,
the parts a person most needs to judge are the new rule and the
examples that illustrate it, not the plumbing. Keeping require-
ments in their own file, with scenarios that cite them and a check
that keeps the two in step, puts the intent of a change in a small,
separate diff. We claim this lowers the reviewer’s load; we have
not yet measured it. UC11 is the agent’s problem: every session
starts cold, and guidance that grows to compensate ends up

Table 6 Twelve use cases in six families. The last two columns give the rung at which the receivers asserted the invariant at commit 8c4a74a and now.

	Situation	Invariant at stake	Then	Now
<i>Placement: sprawl spreads one rule across many places</i>				
UC1	A memoryless agent adds a feature	every part lands in the layer that owns it	L3	L4
UC2	A side effect is added to a use case	use cases publish facts; reactions subscribe	L2	L4
UC3	A business rule drifts to the edge	routes translate; they never decide	L0	L4
<i>Contamination: convenience pulls infrastructure into the core</i>				
UC4	A framework type enters the domain	the domain imports only the standard library	L3	L4
UC5	An outside capability leaks into a service	each capability has one owning layer	L3	L4
<i>Contract integrity: the promises the domain makes</i>				
UC6	A new failure mode is introduced	every error has one public translation	L5	L5
UC7	A test double lies	every double passes the real contract	L3	L3
<i>Harness erosion: the protections themselves wear down</i>				
UC8	The agent edits a rule to go green	rules change only with a person's decision	L2	L2
UC9	A dependency is added	new dependencies need a person	L2	L2
<i>Cognitive load: what people and agents can hold in mind</i>				
UC10	A reviewer faces a large AI diff	intent is reviewable apart from code	L1	L3
UC11	An agent starts cold	the declaration stays small	L3	L3
<i>Transfer: taking the approach to a new domain</i>				
UC12	A new domain is embedded	each invariant is placed on the ladder deliberately	L1	L1

unread. Fading keeps the manifest under 500 words even after two new kinds of intent were added.

6.6. Transfer

UC12 asks whether the pattern survives a move to a domain that is not a task board. Two illustrations, not implemented, suggest how. In a double-entry ledger, the invariant that debits equal credits can be stated in EARS as an unwanted behaviour, shown in a posting slice, checked by a property test, and prevented by a transaction type that can only be built from balanced legs. In medication dosing, the invariant that a dose may not exceed a weight-based maximum without a clinician's override can be checked by boundary scenarios and prevented, in part, by types that distinguish milligrams from milligrams per kilogram. In both, the question the ladder asks is the same: how high must this rule climb, given what it costs to get it wrong?

7. Evaluation

We ask three questions. **RQ1**: which mistakes does each rung catch, and which does it miss? **RQ2**: what do the added rungs cost in size and reading? **RQ3**: do the declared and enforced rungs change what coding agents do? Every number below can be regenerated from the repository's `evaluation/` directory (Griffin 2026).

7.1. RQ1: violation seeding

A seeding script plants one canonical mistake at a time in a receiver, runs its single verify command, records the earliest mechanism that objects, and restores the file. It refuses to start unless the receiver is green, and checks that it is green again at the end. The mechanisms, in order of how early they give feedback, are the compiler, an architecture rule, the requirement check, a behavioural test, and nothing. Table 7 lists the fourteen mistakes: twelve structural, and two that break the trace between requirements and scenarios.

Figure 6 compares the receivers at commit 8c4a74a, before the work described here, with the receivers now. Before, the single gate caught 30 of 36 structural mistakes. The six misses were the same two mistakes in every language: V8, a side effect inlined in the service, and V10, a business rule placed in the route. That the gap was identical across three languages and three enforcement styles is itself a result: it belonged to the capability, not to any receiver. Both mistakes preserve behaviour, so no behavioural test could have caught them; only a structural rule could. After two such rules were added, and the requirement check with its two seeds, the gate catches all 42 mistakes. The new routes rule also moved V7 from a behavioural test to an architecture rule in Python and TypeScript, which is earlier feedback for the same mistake. Application code did not change between the two runs; only tests, manifests and requirement

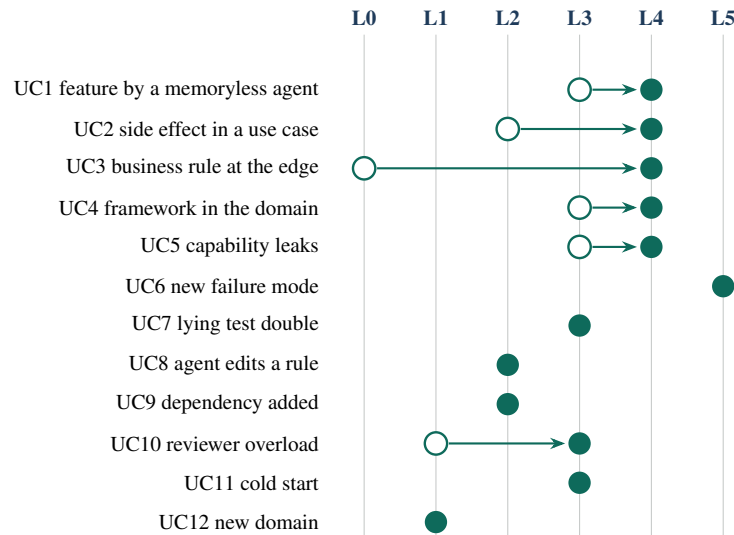


Figure 5 Where each use case sits on the ladder in the receivers, at commit 8c4a74a (hollow) and now (filled). UC6 is at L5 in Java and TypeScript and at L3 in Python.

files did.

7.2. RQ2: cost

Table 8 reports size. Application code is unchanged. Test code grew by between 31 and 89 lines per receiver, and each manifest by between 80 and 83 words, remaining under the 500 words the plan allows. The added rungs therefore cost about one screen of test code and one paragraph of prose per receiver, most of it the requirement check and the ids on existing tests.

7.3. RQ3: a pilot with coding agents

Eighteen coding agents, each starting with no memory, implemented the same feature request, task deletion, in copies of the receivers at commit 8c4a74a. Each copy offered one, two or three of the carriers: the exemplar only (L1), the exemplar and the manifest (L1–L2), or the repository as published (L1–L3). The design crossed three languages, three conditions and two models, with one run per cell. A hidden acceptance test of four cases, never shown to an agent, judged the result, and a scripted audit fixed in advance of inspection recorded where each agent placed each part of the change.

All eighteen runs delivered a passing suite and passed every acceptance case (Table 9). Two departed from the conventions, both by the smaller model in the exemplar-only condition. One, in Java, logged the deletion inside the service instead of publishing an event: the V8 mistake, which no rule caught at the time. The deletion request’s last clause, “whenever a task is deleted the team must be told”, is an event-driven requirement in EARS terms, and the mapping in Table 3 places it in an event handler. The other, in Python, did not add the new operation to the in-memory double, which only the type checker in the full rule set notices. With one run per cell these results are suggestive, not significant. The pilot has not yet been repeated against the receivers as they stand now, with the deletion request written in EARS; that is the next experiment.

8. Discussion

8.1. Lean explains the ladder

Lean thinking faced a version of the problem this paper faces: work done quickly by many hands, more of it than anyone could inspect, where a defect that travelled downstream cost far more to fix than one caught where it was made (Ohno 1988; Womack & Jones 1996). Its answer was to build the specification into the work itself. This section argues that the Assertion Ladder is that answer applied to a repository and a memoryless contributor, through four lean ideas: value, standard work, the value stream and root cause analysis. Table 10 maps each to the receivers and to the evidence, and the section closes with the places where the analogy stops.

Value Lean asks first what the customer values, and the ladder adds the domain to the answer: a change is valuable only if it works and leaves the domain’s rules in one place. Sprawl is then easy to name as waste. It is work that looks finished but will be redone, and the rework falls on whoever next tries to change the rule. The same lens explains the force most easily forgotten, ease of use. A paragraph of prose that repeats what a test enforces is read by every contributor in every session and adds nothing the gate does not already guarantee; for a contributor that relearns everything each time, that reading is the dominant cost. Fading removes it. The receivers show both halves: application code fell from 637, 729 and 957 lines to 283, 223 and 242 when containers, mappers and a second entity were removed, and the manifests stayed under 500 words while two new kinds of intent were added.

Standard work Standard work is the centre of the mapping. The worked slice and the feature recipe of Section 4 (state the rule in EARS, let its pattern choose the layer, copy the slice, run the gate) specify the content of a change, its sequence and its expected outcome, which is how Spear and Bowen describe Toyota’s work (Spear & Bowen 1999). The ladder then does

Table 7 The fourteen seeded mistakes, and the mechanism that caught each in Java / Python / TypeScript now. C compiler, A architecture rule, R requirement check, B behavioural test.

	Mistake	J / P / T
V1	The API bypasses the service and uses a repository	A / A / A
V2	A service signals failure with an HTTP concept	A / A / B
V3	A service issues SQL	A / A / A
V4	A component constructs its own collaborator	A / A / A
V5	A service reads the environment	A / A / A
V6	A new domain error has no HTTP status	C / A / C
V7	A route catches a domain error and answers ad hoc	B / A / A
V8	A side effect is inlined in the service	A / A / A
V9	The test double drifts from the real repository	B / B / B
V10	A business rule is placed in the route	A / A / A
V11	A setting is hard-coded in the service	B / B / B
V12	A framework type enters the domain	A / A / A
V13	A requirement has no scenario	R / R / R
V14	A scenario cites no requirement	R / R / R

Table 8 Size of the receivers at commit 8c4a74a and now (evaluation/metrics*.json).

Receiver	Application lines		Test lines		Manifest words	
	then	now	then	now	then	now
Java	283	283	287	376	397	480
Python	223	223	193	254	347	427
TypeScript	242	242	193	224	388	468

what they found Toyota does: it attaches a test to the standard, so that a departure shows up at once. The lower rungs show and tell the standard; the upper rungs make it self-checking. Every pilot run followed the slice's layering, and the two departures the pilot recorded both came from the condition with the slice alone and no written standard. Standard work is also owned. In lean, the people who do the work improve the standard; in the receivers, an agent follows the standard, and the HUMAN BOUNDARY requires a person's decision before the rules or the manifest change. That division is deliberate, and we return to it below.

The value stream Figure 7 draws the value stream of one change: written by an agent, verified by the single gate, reviewed by a person, merged and released. Each rung decides where along that stream a broken rule is found. At L0 only a reviewer who reads closely can see it, and cognitive erosion is precisely the loss of that reading. At L1 and L2 the standard makes the mistake less likely but finds nothing. At L3 and L4 the gate finds it before any person has spent time on the change, and at L5 it cannot be written. The seeding results show the

shift. At commit 8c4a74a, V8 and V10 passed the gate in all three receivers and would have reached review; now both are stopped at the gate by an architecture rule. V7 moved earlier within the gate, from a behavioural test to an architecture rule, in Python and TypeScript, and V6 was already refused by the compiler in Java and TypeScript. We have no timing data for the stream, so we claim only the position at which a defect is found, not the time saved. The lean argument is that position matters: a defect found at the gate goes back to the contributor who made it while the change is still open, whereas one found in review has already cost a person's attention and, if missed, becomes the rework that sprawl describes.

Root cause analysis The first seeding run is a small worked example of lean problem solving. The problem was concrete: six of 36 seeded mistakes passed the gate. Asking why led to a cause at the level of the ladder, not of any receiver. The misses were the same two mistakes, V8 and V10, in all three languages. Both passed because every behavioural test passed; every behavioural test passed because neither mistake changes what the system does; and nothing else looked, because the structural

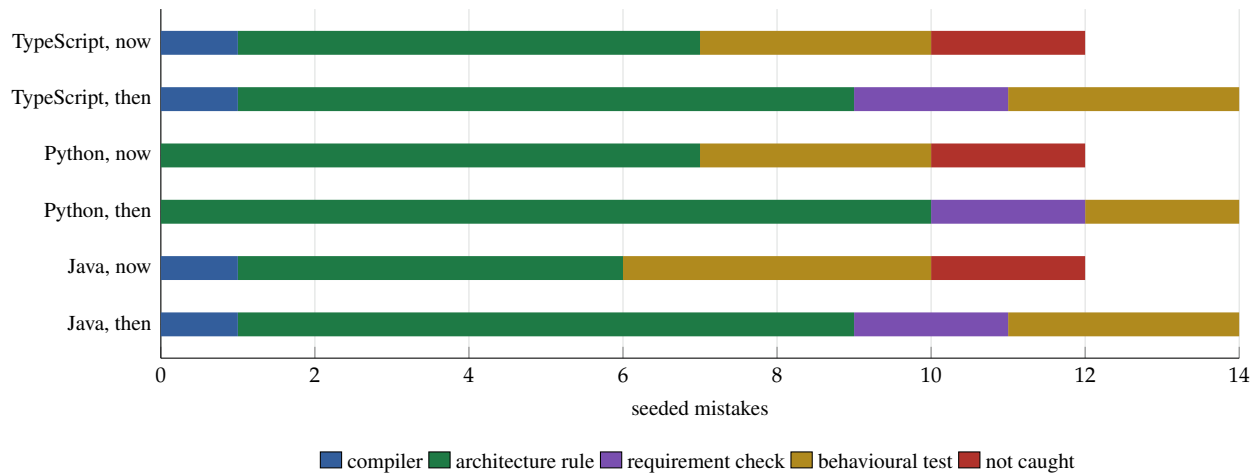


Figure 6 Seeded mistakes by the earliest mechanism that caught them, per receiver, at commit 8c4a74a (12 structural seeds) and now (14 seeds).

Table 9 Pilot results by condition (six runs each; evaluation/agent_experiment/).

Condition	Acceptance passed	Departures from convention	Mean tokens	Mean seconds
Exemplar (L1)	6 / 6	2	61 650	121
+ Manifest (L1–L2)	6 / 6	0	61 990	132
+ Enforced (L1–L3)	6 / 6	0	65 696	151

rules covered which layer may import which, not which layer may log, notify or decide. The root cause was that behaviour cannot see placement and no rule asserted placement for side effects or decisions. The countermeasure was two EXECUTABLE RULES, each with a TEACHING FAILURE naming where the code belongs, which is to say two rules moved from L0 and L2 to L4. The check was a second run, which caught all 42 seeds, including the new requirement seeds. The standard then absorbed the countermeasure, in the plan–do–check–act manner of an A3 report (Sobek II & Smalley 2008). The V9 fixture episode (Section 6) followed the same cycle on a smaller scale, and is the kaizen of a check rather than of the code. What lean adds to the usual advice to add a test after a bug is the insistence on the cause: because the same gap appeared in three languages, the countermeasure was a kind of rule every receiver needed, not a patch to one of them.

Supporting mechanisms Three further lean mechanisms map onto the ladder more briefly. The single gate is *jidoka*: the build stops itself on an abnormality, so that no person has to watch every line, and the manifest’s instruction to change the code rather than the test is the rule that a stop is answered by fixing its cause. No pilot run modified a rule. Jidoka’s older name, automation with a human touch, fits a team of people and agents well: the machine detects and stops, the person judges. The TEACHING FAILURE is the *andon* signal, which shows where the problem is and why; the HUMAN BOUNDARY is the andon cord, the right and the duty to stop and ask a person before changing a dependency, a rule or a schema. The

UNREPRESENTABLE MISTAKE is *poka-yoke* of the preventive kind.

Where the analogy stops The mapping is close but not exact, and the differences are instructive. First, an agent does not learn across shifts. A factory worker is taught the standard, practises it and in time no longer needs the sheet; an agent is taught afresh every session. The cost of showing and telling is therefore paid every time, which is the strongest reason to move a rule from prose into a check, where following it costs no reading. For the same reason, improvement cannot happen inside the agent. Whatever an agent notices in a session is lost unless it is written into the repository, so reflection and kaizen run through people and the files they own. Second, lean’s respect for people becomes the HUMAN BOUNDARY. In lean the people who do the work own the standard. Here much of the work is done by agents, so people keep what needs judgement: what counts as value, how high a rule should climb, and when a check is wrong. The risk to avoid is the one lean warned against, reducing people to inspectors of someone else’s output; the ladder gives them the standard and its checks to own instead of every line to read. Third, the fixture is made of the same material as the work. A worker cannot redesign the jig in the middle of a shift, but an agent can edit a test, which is why changes to the rules sit behind the HUMAN BOUNDARY, and why use case UC8 remains at L2 until repository settings enforce it. Fourth, the ladder can itself become waste. A check is cheap to run but not free to maintain, and a rule encoded in types far more elaborate than its risk warrants is over-processing. Reinertsen’s caution

Table 10 Lean ideas, the rungs and patterns that carry them, and where they appear in the receivers. The first four rows are the argument; the last three are supporting mechanisms.

Lean idea	Rung or pattern	In the receivers	Evidence
Value, and waste removed	fading; SHORT MANIFEST	no DI container, no ORM, one entity; no prose for what a test checks	637 / 729 / 957 → 283 / 223 / 242 lines; manifests 480 / 427 / 468 words
Standard work	L1 WORKED SLICE, L2 SHORT MANIFEST, with L3 as its built-in test	the slice and the feature recipe, followed by every contributor and changed only by people	every pilot run followed the slice’s layering
Value stream: defects found earlier	L3–L5	V8 and V10 now stopped at the gate, not left to review; V7 moved from a behavioural test to an architecture rule in Python and TypeScript	V8 and V10 passed the gate in all three receivers at 8c4a74a; caught now
Root cause analysis and kaizen	EXECUTABLE RULE, TEACHING FAILURE; HONEST DOUBLE	the misses traced to one cause and closed by two rules; the V9 fixture corrected	30 / 36 → 42 / 42; V9 caught in all three receivers
Jidoka (the line stops itself)	SINGLE GATE	one verify command; “change the code, not the test”	no pilot run modified a rule (18 / 18)
Andon (signal and cord)	L4 TEACHING FAILURE; HUMAN BOUNDARY	failures name the rule and the fix; “ask a human first” for dependencies, rules and schema	message on every rule; boundary declared (L2), not yet enforced
Poka-yoke (mistake-proofing)	L5 UNREPRESENTABLE MISTAKE	exhaustive error mapping that will not compile when a case is missing	V6 caught by the compiler in Java and TypeScript

applies (Reinertsen 2009), and so do the findings of Staats and colleagues on uncertain knowledge work (Staats et al. 2011): standardise what should repeat and leave the rest free. The ladder applies lean to a team’s conventions, not to its design.

8.2. Domain sanctity against ease of use

A repository can be made arbitrarily hard to damage by making it arbitrarily hard to change. The ladder resists this in two ways. Each rule climbs only as far as its risk warrants, so the cost of a rung is paid only where it buys something; in lean terms, this guards against over-processing. And the ladder is compressed from below as it grows from above: when the receivers gained a requirement file, a scenario convention and four new rules, their manifests grew by one paragraph each and stayed under 500 words, so the written standard stayed short enough to be read in every session. The evidence that ease of use survived is modest but concrete: in the pilot, agents given the full rule set finished in 151 seconds on average against 121 for the exemplar alone, and used about 6% more tokens, for a condition in which none departed from the conventions.

8.3. Where each kind of intent stops

Each kind of intent sees some mistakes and is blind to others, which is why the three are complementary rather than redundant. Behaviour cannot see placement: V8 and V10 pass every scenario because what the system does is unchanged. Structure cannot see meaning: an architecture rule will not notice that a task may now skip a step. Requirements cannot execute: with-

out the closure check that ties them to scenarios, they decay into prose, the failure the manifest’s own design avoids. The ladder’s contribution is to put all three on the same scale, so that a team can ask of any rule, in any of the three, how strongly it is asserted.

8.4. Threats to validity

Construct Seeded mistakes are canonical, not observed; real agents may make mistakes the fourteen do not represent. The TypeScript architecture rules match source text, not syntax, and could be evaded by code written to avoid them.

Internal The authors built the receivers, the seeds and the rules together with an AI coding agent, in the collaborative mode the paper studies. Rules were added after the misses they close were known; the second seeding run therefore shows that the gap can be closed, not that the rules generalise to unseen mistakes.

External One small domain, three mainstream stacks and two models from one vendor limit generality. The pilot ran one agent per cell and has not yet been repeated against the current receivers.

Conclusion Seeding results are deterministic and reproducible from the repository. Pilot results are suggestive only.

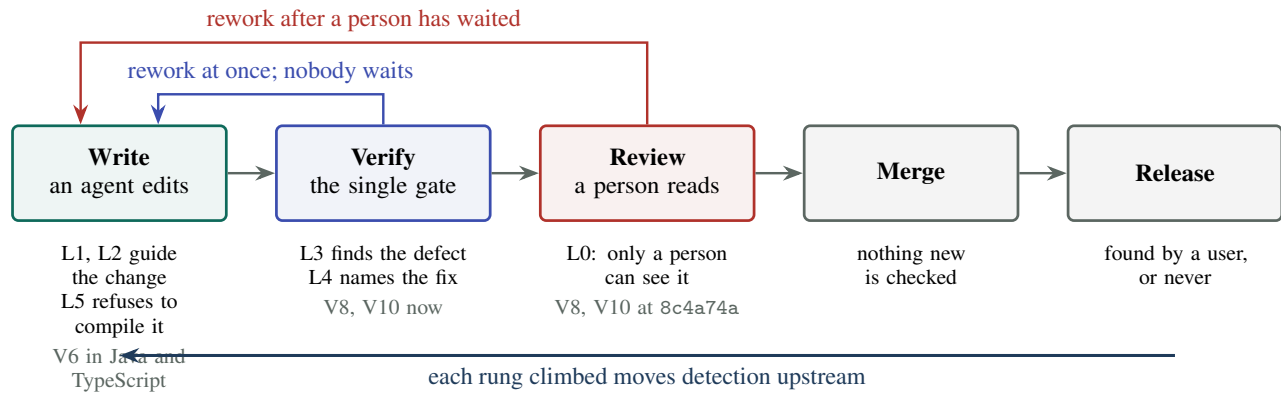


Figure 7 The value stream of one change. Each rung moves the point at which a broken rule is found further upstream. At commit 8c4a74a, V8 and V10 passed the gate and could be seen only in review; they are now stopped at the gate. The figure shows position only; we have no timing data.

9. Related work

The Assertion Ladder draws on six bodies of work that rarely cite one another: architecture conformance, behaviour and requirement specification, studies of coding agents, constraints on generated code, the psychology of learning, and the pattern form. Lean production is treated in Section 8. This section reviews each body of work and says how the ladder builds on it. Each theme ends with the gap in current practice that the ladder proposes to standardise. The gaps are collected at the end as G1 to G5.

9.1. Architecture conformance, erosion and drift

The gap between intended and implemented architecture has been studied for more than thirty years. Perry and Wolf separated *drift*, a growing insensitivity to the architecture, from *erosion*, outright violation of it (D. E. Perry & Wolf 1992). Lehman's laws state that a changing system grows more complex unless work is done to stop it (Lehman 1980). The debt metaphor gives that work a price: shortcuts are borrowed against future change and repaid with interest (Cunningham 1992; Kruchten et al. 2012). Two surveys organise the response. De Silva and Balasubramaniam classify erosion-control techniques into those that minimise, prevent or repair erosion (de Silva & Balasubramaniam 2012). Li et al. map the symptoms, causes and consequences of erosion reported across the literature (R. Li et al. 2022).

Conformance checking is the best-studied remedy. Reflexion models compare a high-level model that an engineer proposes with a model extracted from the source, and report convergences, divergences and absences (Murphy et al. 2001). Dependency constraint languages state the allowed and forbidden dependencies between modules and check them statically (Terra & Valente 2009). Passos et al. compare these and related approaches on one system (Passos et al. 2010). Practice has since moved the checks into the build. Architecture fitness functions make an architectural property executable (Ford et al. 2017). ArchUnit expresses dependency rules for Java as ordinary unit tests (TNG Technology Consulting n.d.), Import Linter enforces layer and

independence contracts between Python modules (Seddon n.d.), and dependency-cruiser validates JavaScript and TypeScript imports against rules in a configuration file (*dependency-cruiser: Validate and Visualise Dependencies, with Your Rules* n.d.).

The receivers use this tradition directly: the EXECUTABLE RULE at L3 is a conformance check. The ladder adds two things the tradition does not address. It asks which rules deserve a check at all, since a check has a cost and some rules are better shown or told. It also asks that a failing check teach the fix, which matters when the reader of the failure is an agent that will act on it immediately. What is missing is a principle for choosing which rules to check and which to document, and a requirement that a failing check explain itself (G3).

9.2. Specification by example, BDD and requirements

Behaviour-driven development grew out of test-driven development (Beck 2003). It names tests as behaviours in the language of the domain and writes them as given, when and then (North 2006). Specification by example treats a set of such examples as the living documentation of a system (Adzic 2011), and Gherkin with Cucumber makes them executable as plain-text scenarios (Wynne & Hellesøy 2012). Solis and Wang identified the characteristics that define BDD, among them a ubiquitous language, plain-text scenario templates and automated acceptance tests (Solis & Wang 2011). The research record is uneven. Binamungu et al. found that practitioners struggle to keep large scenario suites consistent and free of duplication as they grow (Binamungu et al. 2018). A later mapping study of 166 papers found little industrial evaluation and few metrics for the quality of BDD specifications (Binamungu & Maro 2023).

EARS constrains natural-language requirements to a few templates in order to remove ambiguity (Mavin et al. 2009). Its authors report ten years of adoption across industrial domains (Mavin & Wilkinson 2019). Requirements traceability has a longer history. Gotel and Finkelstein showed that most traceability problems arise before the requirement is written, in the loss of its origin and rationale (Gotel & Finkelstein 1994). Bjarnason et al. studied six companies and reported recurring challenges in keeping requirements aligned with testing, along

with the practices that address them (Bjarnason et al. 2014).

The receivers take the scenario form but not the Gherkin runner. Scenarios are ordinary tests, named in given, when and then, that cite a requirement id. The ladder then adds what the literature treats as an open maintenance problem: a CLOSURE CHECK that fails the build when a requirement has no scenario or a scenario cites no requirement. It also uses the EARS keyword as a hint about placement, a use its designers did not intend. What is missing is an executable closure between requirements and scenarios in repositories that agents edit, so that traceability is checked on every change rather than audited occasionally (G4).

9.3. Coding agents, their harnesses and their context files

Code generation moved from completing functions (M. Chen et al. 2021) to resolving real issues (Jimenez et al. 2024). The step depended on the harness. SWE-agent showed that an agent-computer interface designed for the model changes what the model can achieve (Yang et al. 2024). AutoCodeRover added code search over the syntax tree and test-based fault localisation (Zhang et al. 2024), and OpenHands provides an open platform for such agents (Wang et al. 2025). These benchmarks and systems measure the agent. The repository is held fixed and treated as the problem to be solved.

Empirical studies of assisted programming give a mixed picture. A controlled experiment found developers with Copilot finished a task 55.8% faster (Peng et al. 2023). A randomised trial with experienced open-source maintainers found that AI tools made them 19% slower (J. Becker et al. 2025). The correctness of suggestions varies by language (Nguyen & Nadi 2022), and security studies show that insecure code passes unnoticed (Pearce et al. 2022; N. Perry et al. 2023). Studies of the process describe a shift from writing to reviewing (Vaithilingam et al. 2022; Barke et al. 2023). At the scale of repositories, an industry report found rising code churn and copy-pasted code and falling moved code as assistants spread (Harding & Kloster 2024). A difference-in-differences study of projects that adopted Cursor found a short rise in velocity and a lasting rise in static-analysis warnings and complexity (He et al. 2026). In a dataset of 456,000 pull requests written by agents, those pull requests were accepted less often than human ones but reviewed much faster (H. Li et al. 2025). These are the conditions under which sprawl and cognitive erosion arise.

Agent context files are the newest object of study. AGENTS.md proposes one Markdown file per repository, read at the start of every session (AGENTS.md: A Simple, Open Format for Guiding Coding Agents n.d.). Studies of such files find shallow structure and mostly operational content (Chatlatanagulchai, Thonglek, et al. 2025), no settled structure across 466 open-source projects (Mohsenimofidi et al. 2026), and, across 2,303 files, frequent guidance on testing and architecture but little on security or performance (Chatlatanagulchai, Li, et al. 2025). Evidence on their effect is mixed. One study associates AGENTS.md files with shorter agent runtimes and fewer tokens at the same completion rate (Lulla et al. 2026). Another finds that context files do not generally improve task success

and raise inference cost by over 20% (Gloaguen et al. 2026). Work on generated tests shows that models can write useful unit tests (Schäfer et al. 2024), and a review of LLMs in software architecture finds few studies of architecture-aware generation (Schmid et al. 2025).

The ladder places a context file at L2, one rung among six, and treats its growth as a cost. The mixed evidence on context files is what the ladder predicts. Prose that repeats what a test enforces adds cost without adding protection, and prose that states what no test enforces is easily ignored. The receivers keep the manifest short by removing a rule from it once a check covers it. What is missing is a link between the instruction file and the checks, so that each told rule is either enforced or deliberately left to judgement (G2). A second gap follows from the benchmarks. SWE-bench and its successors measure agents against fixed repositories. What is missing is a measure of the repository, one that asks whether its harness catches a known set of canonical mistakes whoever makes them (G5). Violation seeding (Section 7) is a first form of such a measure.

9.4. Guardrails and constraints on generated code

A second line of work constrains the model rather than the repository. Constrained decoding restricts generation to programs that parse or type-check. Synchromesh enforces grammar and semantic constraints during decoding (Poesia et al. 2022). Monitor-guided decoding consults a static analysis of the repository so that generated code refers only to names that exist (Agrawal et al. 2023). Type-constrained decoding for TypeScript more than halves compilation errors (Mündler et al. 2025). Other work places feedback in a loop after generation: models improve when they see execution results and explain their own code (X. Chen et al. 2024). Tests can also serve as the specification. Supplying tests with a problem statement raises the rate at which models solve it (Mathews & Nagappan 2024), and generating tests first to clarify intent helps both the model and the programmer who checks its output (Fakhoury et al. 2024).

These techniques sit at particular rungs. Constrained decoding is L5 applied at the model, and feedback loops are L3 and L4 applied at the harness. The ladder places the same assertions in the repository, where they survive a change of model, vendor or harness and bind people as well as agents. The two approaches are complementary. A repository whose rules are executable gives a feedback loop something to run, and a type-level assertion gives constrained decoding something to enforce. What is missing is a shared vocabulary for how strongly a rule is asserted, so that a team can say which of its rules are shown, told, checked, explained or prevented, whatever tools do the asserting (G1).

9.5. Pedagogy and cognitive load in software engineering

Cognitive load theory holds that working memory is small and that instruction should not waste it (Sweller 1988). Worked examples are one of its most robust findings (Sweller & Cooper 1985), and minimal guidance is less effective than direct instruction for people new to a domain (Kirschner et al. 2006).

Scaffolding gives support that is later withdrawn (Wood et al. 1976), and fading turns worked examples gradually into problems (Renkl & Atkinson 2003). The expertise reversal effect warns that guidance which helps a novice can hinder an expert (Kalyuga et al. 2003). Formative feedback is most useful when it is specific and timely (Shute 2008), and programming error messages have long been criticised for failing to say what to do (B. A. Becker et al. 2019).

Software engineering has studied the same limits. Program comprehension research shows how much of development is spent understanding code rather than writing it (Siegmund 2016), and a mapping study catalogues how the cognitive load of developers is measured (Gonçalves et al. 2019). In code review, understanding the change is the main challenge (Bacchelli & Bird 2013), and working memory capacity is associated with review effectiveness (Baum et al. 2019).

The ladder applies these findings to a new kind of learner. An agent is expert in general and a novice in the particular repository, so the ladder gives it direct instruction and worked examples for the particular (L1, L2) and fades the prose once a check exists, as the expertise reversal effect would advise. TEACHING FAILURE is formative feedback written into the checks. The reviewer's load is reduced by moving rules out of the diff and into the gate. What is missing, again, is a shared scale on which these teaching moves are named and graded, so that repository design can draw on instructional design deliberately (G1).

9.6. Patterns and pattern languages

A pattern names a problem in a context, the forces that act on it and a resolution (Alexander et al. 1977). The form reached software through the Gang of Four (Gamma et al. 1994) and the POSA series (Buschmann et al. 1996), whose fifth volume examines how patterns combine into sequences and languages (Buschmann et al. 2007). Enterprise application patterns catalogue layering and the service layer used by the receivers (Fowler 2002). Test patterns are closest to the rung patterns. Meszaros catalogues test doubles, including the fake object that stands in for a real dependency (Meszaros 2007). HONEST DOUBLE adds a contract that both the fake and the real implementation must pass.

The Assertion Ladder is a pattern in this classic form with nine smaller patterns beneath it. Its forces are specific to repositories shared with agents: memoryless contributors, review that cannot keep pace, and the cost of every assertion. Existing catalogues name the solutions the rung patterns use, but none grades them against one another.

9.7. Our earlier work

Our earlier paper proposed a repository standard in which conventions are exemplified by one small vertical slice, declared in a short manifest and enforced by architecture rules written as tests (Griffin et al. 2026). In the vocabulary of this paper, that standard binds L1, L2 and L3 for structure alone. This paper generalises it in three ways. It adds the rungs above and below (L0, L4 and L5) and so grades assertion by strength. It extends the same rungs from structure to behaviour and requirements.

And it reports a second seeding run with fourteen canonical mistakes rather than twelve. The new run closes the two misses that the first run found in every language, which motivated the behavioural and requirement rungs.

Gaps The review leaves five gaps in current practice. The Assertion Ladder proposes a standard for each.

G1 There is no shared vocabulary for how strongly a rule is asserted.

G2 Agent instruction files are unvalidated prose with no link to enforcement.

G3 Conformance tools do not say which rules to check rather than document, or require a failure to name its fix.

G4 Agent-facing repositories have no executable closure between requirements and scenarios.

G5 No benchmark asks whether a repository's harness catches canonical mistakes; SWE-bench measures agents, not repositories.

10. Conclusion

Coding agents make a domain's rules easier to break and harder to notice breaking. The Assertion Ladder answers by asking, for each rule, how strongly the repository asserts it: shown, told, checked, explained or prevented. Structure, behaviour and requirements turn out to be three loads on the same ladder, and the pattern of an EARS requirement suggests where its code belongs. A greenfield receiver in TypeScript, with equivalent receivers in Java and Python, binds every rung in its language's idiom; seeding fourteen canonical mistakes into each, the single verify command catches all 42, where it caught 30 of 36 structural mistakes before the behavioural and requirement rungs and two new structural rules were added. The cost was about a screen of tests and a paragraph of prose per receiver. The Tutors retrofit shows the same rungs established in a production codebase that agents were already changing, reaching the checked and explained rungs wherever a check could be written, adding ratchets so that legacy violations could only shrink, and extracting its release checks into a harness that the change under review cannot weaken.

We therefore propose the ladder as a standard for repositories shared with coding agents, in answer to the five gaps of Section 9: a shared vocabulary of rungs; instruction files that state only what is not yet checked; checks chosen by risk and written to teach; requirements closed against scenarios; and violation seeding as a benchmark of the repository itself rather than of the agent that works in it.

Lean thinking explains why this works. It counts as value only what the customer, and here the domain, needs, and treats the rest as waste. It makes the best current method standard work that everyone follows and the team improves, it asks where along the value stream a defect is found, and it answers each miss by asking why until the process can change. The ladder is standard work, with its tests built in, for a contributor who cannot remember it; each rung it climbs moves detection further upstream; and the two seeding runs are one turn of lean's improvement cycle, from a miss to its root cause to a counter-measure that was measured again.

Four directions follow. The pilot should be repeated against the current receivers, with more runs per cell and the feature request written in EARS, to test whether stated requirements improve placement. The harness-erosion use cases should be raised from L2 to L3 by protecting the rule files themselves. The extracted release harness should be adopted by a second product, to test how much of it is generic. And the reviewer's load, which we claim the ladder reduces by separating intent from code, should be measured.

Availability The receivers, the seeding script, the pilot's workspaces, audit and diffs, and this paper's sources are public (Griffin 2026), as are Tutors and its release harness (Tutors contributors 2026a,b). Each receiver can be read on its own, starting from its AGENTS.md.

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References

Adzic, G. (2011). *Specification by example: How successful teams deliver the right software*. Manning.

AGENTS.md: A simple, open format for guiding coding agents. (n.d.). <https://agents.md>. (Accessed 2026)

Agrawal, L. A., Kanade, A., Goyal, N., Lahiri, S., & Rajamani, S. (2023). Monitor-guided decoding of code LMs with static analysis of repository context. In *Advances in neural information processing systems 36 (neurips)* (pp. 32270–32298).

Alexander, C., Ishikawa, S., & Silverstein, M. (1977). *A pattern language: Towns, buildings, construction*. Oxford University Press.

Ali, N. B., Petersen, K., & Schneider, K. (2016). FLOW-assisted value stream mapping in the early phases of large-scale software development. *Journal of Systems and Software*, 111, 213–227. doi: 10.1016/j.jss.2015.10.013

Bacchelli, A., & Bird, C. (2013). Expectations, outcomes, and challenges of modern code review. In *35th international conference on software engineering (icse)* (pp. 712–721). IEEE. doi: 10.1109/ICSE.2013.6606617

Barke, S., James, M. B., & Polikarpova, N. (2023). Grounded copilot: How programmers interact with code-generating models. *Proceedings of the ACM on Programming Languages*, 7(OOPSLA1). doi: 10.1145/3586030

Baum, T., Schneider, K., & Bacchelli, A. (2019). Associating working memory capacity and code change ordering with code review performance. *Empirical Software Engineering*, 24(4), 1762–1798. doi: 10.1007/s10664-018-9676-8

Beck, K. (2003). *Test-driven development: By example*. Addison-Wesley.

Becker, B. A., Denny, P., Pettit, R., Bouchard, D., Bouvier, D. J., Harrington, B., ... Prather, J. (2019). Compiler error messages considered unhelpful: The landscape of text-based programming error message research. In *Proceedings of the working group reports on innovation and technology in computer science education (iticse-wgr)* (pp. 177–210). ACM. doi: 10.1145/3344429.3372508

Becker, J., Rush, N., Barnes, E., & Rein, D. (2025). *Measuring the impact of early-2025 AI on experienced open-source developer productivity*. arXiv:2507.09089. doi: 10.48550/arXiv.2507.09089

Binamungu, L. P., Embury, S. M., & Konstantinou, N. (2018). Maintaining behaviour driven development specifications: Challenges and opportunities. In *Ieee 25th international conference on software analysis, evolution and reengineering (saner)* (pp. 175–184). IEEE. doi: 10.1109/SANER.2018.8330207

Binamungu, L. P., & Maro, S. (2023). Behaviour driven development: A systematic mapping study. *Journal of Systems and Software*, 203, 111749. doi: 10.1016/j.jss.2023.111749

Bjarnason, E., Runeson, P., Borg, M., Unterkalmsteiner, M., Engström, E., Regnell, B., ... Feldt, R. (2014). Challenges and practices in aligning requirements with verification and validation: A case study of six companies. *Empirical Software Engineering*, 19(6), 1809–1855. doi: 10.1007/s10664-013-9263-y

Buschmann, F., Henney, K., & Schmidt, D. C. (2007). *Pattern-oriented software architecture, volume 5: On patterns and pattern languages*. Wiley.

Buschmann, F., Meunier, R., Rohnert, H., Sommerlad, P., & Stal, M. (1996). *Pattern-oriented software architecture, volume 1: A system of patterns*. Wiley.

Chatlatanagulchai, W., Li, H., Kashiwa, Y., Reid, B., Thonglek, K., Leelaprute, P., ... Iida, H. (2025). *Agent READMEs: An empirical study of context files for agentic coding*. arXiv:2511.12884. doi: 10.48550/arXiv.2511.12884

Chatlatanagulchai, W., Thonglek, K., Reid, B., Kashiwa, Y., Leelaprute, P., Rungsawang, A., ... Iida, H. (2025). On the use of agentic coding manifests: An empirical study of Claude Code. In *Proceedings of the 26th international conference on product-focused software process improvement (profes)*. (arXiv:2509.14744)

Chen, M., et al. (2021). *Evaluating large language models trained on code*. arXiv:2107.03374.

Chen, X., Lin, M., Schärli, N., & Zhou, D. (2024). Teaching large language models to self-debug. In *International conference on learning representations (iclr)*.

Cockburn, A. (2005). *Hexagonal architecture*. <https://alistair.cockburn.us/hexagonal-architecture/>.

Cunningham, W. (1992). The WyCash portfolio management system. In *Addendum to the proceedings on object-oriented programming systems, languages, and applications (oopsla '92)* (pp. 29–30). ACM. doi: 10.1145/157709.157715

dependency-cruiser: Validate and visualise dependencies, with your rules. (n.d.). <https://github.com/sverweij/dependency-cruiser>. (Accessed September 2026)

de Silva, L., & Balasubramaniam, D. (2012). Controlling software architecture erosion: A survey. *Journal of Systems and Software*, 85(1), 132–151. doi: 10.1016/j.jss.2011.07.036

Dijkstra, E. W. (1968). The structure of the “THE”-multiprogramming system. *Communications of the ACM*, 11(5), 341–346.

Evans, E. (2003). *Domain-driven design: Tackling complexity*

- in the heart of software. Addison-Wesley.
- Fakhoury, S., Naik, A., Sakkas, G., Chakraborty, S., & Lahiri, S. K. (2024). LLM-based test-driven interactive code generation: User study and empirical evaluation. *IEEE Transactions on Software Engineering*, 50(9), 2254–2268. doi: 10.1109/TSE.2024.3428972
- Ford, N., Parsons, R., & Kua, P. (2017). *Building evolutionary architectures: Support constant change*. O'Reilly.
- Forsgren, N., Humble, J., & Kim, G. (2018). *Accelerate: The science of lean software and DevOps: Building and scaling high performing technology organizations*. IT Revolution.
- Fowler, M. (2002). *Patterns of enterprise application architecture*. Addison-Wesley.
- Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design patterns: Elements of reusable object-oriented software*. Addison-Wesley.
- Gloaguen, T., Mündler, N., Müller, M., Raychev, V., & Vechev, M. (2026). Evaluating AGENTS.md: Are repository-level context files helpful for coding agents? In *Iclr 2026 workshop on memory for llm-based agentic systems (memagents)*. (arXiv:2602.11988)
- Gonçales, L., Farias, K., da Silva, B., & Fessler, J. (2019). Measuring the cognitive load of software developers: A systematic mapping study. In *Ieee/acm 27th international conference on program comprehension (icpc)* (pp. 42–52). IEEE. doi: 10.1109/ICPC.2019.00018
- Gotel, O. C. Z., & Finkelstein, A. C. W. (1994). An analysis of the requirements traceability problem. In *Proceedings of the ieee international conference on requirements engineering (icre)* (pp. 94–101). IEEE. doi: 10.1109/ICRE.1994.292398
- Griffin, L. (2026). *AI reference projects: taskboard in Java, Python and TypeScript*. <https://github.com/lgriffin/AI-Reference-Projects>.
- Griffin, L., et al. (2026). *Convention over configuration for coding agents: A repository standard of default architectural patterns for human-AI software development*. Submitted to the Journal of Object Technology.
- Harding, W., & Kloster, M. (2024, January). *Coding on Copilot: 2023 data suggests downward pressure on code quality* (Tech. Rep.). GitClear. Retrieved from https://www.gitclear.com/coding_on_copilot_data_shows_ais_downward_pressure_on_code_quality
- He, H., Miller, C., Agarwal, S., Kästner, C., & Vasilescu, B. (2026). Speed at the cost of quality: How Cursor AI increases short-term velocity and long-term complexity in open-source projects. In *Proceedings of the 23rd international conference on mining software repositories (msr)*. (arXiv:2511.04427)
- Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-bench: Can language models resolve real-world GitHub issues? In *International conference on learning representations (iclr)*.
- Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, 38(1), 23–31. doi: 10.1207/S15326985EP3801_4
- Khurum, M., Petersen, K., & Gorschek, T. (2014). Extending value stream mapping through waste definition beyond customer perspective. *Journal of Software: Evolution and Process*, 26, 1074–1105. doi: 10.1002/smr.1647
- Kirschner, P. A., Sweller, J., & Clark, R. E. (2006). Why minimal guidance during instruction does not work: An analysis of the failure of constructivist, discovery, problem-based, experiential, and inquiry-based teaching. *Educational Psychologist*, 41(2), 75–86. doi: 10.1207/s15326985ep4102_1
- Kruchten, P., Nord, R. L., & Ozkaya, I. (2012). Technical debt: From metaphor to theory and practice. *IEEE Software*, 29(6), 18–21. doi: 10.1109/MS.2012.167
- Lehman, M. M. (1980). Programs, life cycles, and laws of software evolution. *Proceedings of the IEEE*, 68(9), 1060–1076. doi: 10.1109/PROC.1980.11805
- Li, H., Zhang, H., & Hassan, A. E. (2025). *The rise of AI teammates in software engineering (SE) 3.0: How autonomous coding agents are reshaping software engineering*. arXiv:2507.15003. doi: 10.48550/arXiv.2507.15003
- Li, R., Liang, P., Soliman, M., & Avgeriou, P. (2022). Understanding software architecture erosion: A systematic mapping study. *Journal of Software: Evolution and Process*, 34(3), e2423. doi: 10.1002/smr.2423
- Liker, J. K. (2004). *The Toyota way: 14 management principles from the world's greatest manufacturer*. McGraw-Hill.
- Lulla, J. L., Mohsenimofidi, S., Galster, M., Zhang, J. M., Baltes, S., & Treude, C. (2026). *On the impact of AGENTS.md files on the efficiency of AI coding agents*. arXiv:2601.20404. doi: 10.48550/arXiv.2601.20404
- Martin, R. C. (2017). *Clean architecture: A craftsman's guide to software structure and design*. Prentice Hall.
- Mathews, N. S., & Nagappan, M. (2024). Test-driven development and LLM-based code generation. In *Proceedings of the 39th ieee/acm international conference on automated software engineering (ase)* (pp. 1583–1594). ACM. doi: 10.1145/3691620.3695527
- Mavin, A., & Wilkinson, P. (2019). Ten years of EARS. *IEEE Software*, 36(5), 10–14. doi: 10.1109/MS.2019.2921164
- Mavin, A., Wilkinson, P., Harwood, A., & Novak, M. (2009). Easy approach to requirements syntax (EARS). In *Proceedings of the 17th ieee international requirements engineering conference (re'09)* (pp. 317–322). IEEE. doi: 10.1109/RE.2009.9
- Meszaros, G. (2007). *xUnit test patterns: Refactoring test code*. Addison-Wesley.
- Mohsenimofidi, S., Galster, M., Treude, C., & Baltes, S. (2026). Context engineering for AI agents in open-source software. In *Proceedings of the 23rd international conference on mining software repositories (msr)*. (arXiv:2510.21413)
- Mündler, N., He, J., Wang, H., Sen, K., Song, D., & Vechev, M. (2025). Type-constrained code generation with language models. *Proceedings of the ACM on Programming Languages*, 9(PLDI), 601–626. doi: 10.1145/3729274
- Murphy, G. C., Notkin, D., & Sullivan, K. J. (2001). Software reflexion models: Bridging the gap between design and implementation. *IEEE Transactions on Software Engineering*, 27(4), 364–380. doi: 10.1109/32.917525
- Nguyen, N., & Nadi, S. (2022). An empirical evaluation of GitHub Copilot's code suggestions. In *Proceedings of the 19th international conference on mining software repositories*

- (msr). ACM. doi: 10.1145/3524842.3528470
- North, D. (2006). *Introducing BDD*. <https://dannorth.net/introducing-bdd/>. (First published as “Behavior Modification” in Better Software magazine, March 2006)
- Ohno, T. (1988). *Toyota production system: Beyond large-scale production*. Productivity Press.
- Parnas, D. L. (1972). On the criteria to be used in decomposing systems into modules. *Communications of the ACM*, 15(12), 1053–1058.
- Passos, L., Terra, R., Valente, M. T., Diniz, R., & das Chagas Mendonça, N. (2010). Static architecture-conformance checking: An illustrative overview. *IEEE Software*, 27(5), 82–89. doi: 10.1109/MS.2009.117
- Pearce, H., Ahmad, B., Tan, B., Dolan-Gavitt, B., & Karri, R. (2022). Asleep at the keyboard? assessing the security of GitHub Copilot’s code contributions. In *Ieee symposium on security and privacy (s&p)*. doi: 10.1109/SP46214.2022.9833571
- Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). *The impact of AI on developer productivity: Evidence from GitHub Copilot*. arXiv:2302.06590. doi: 10.48550/arXiv.2302.06590
- Perry, D. E., & Wolf, A. L. (1992). Foundations for the study of software architecture. *ACM SIGSOFT Software Engineering Notes*, 17(4), 40–52. doi: 10.1145/141874.141884
- Perry, N., Srivastava, M., Kumar, D., & Boneh, D. (2023). Do users write more insecure code with AI assistants? In *Acm sigsac conference on computer and communications security (ccs)*. doi: 10.1145/3576915.3623157
- Poesia, G., Polozov, A., Le, V., Tiwari, A., Soares, G., Meek, C., & Gulwani, S. (2022). Synchromesh: Reliable code generation from pre-trained language models. In *International conference on learning representations (iclr)*.
- Poppendieck, M., & Poppendieck, T. (2003). *Lean software development: An agile toolkit*. Addison-Wesley.
- Poppendieck, M., & Poppendieck, T. (2006). *Implementing lean software development: From concept to cash*. Addison-Wesley.
- Reinertsen, D. G. (2009). *The principles of product development flow: Second generation lean product development*. Redondo Beach, CA: Celeritas Publishing.
- Renkl, A., & Atkinson, R. K. (2003). Structuring the transition from example study to problem solving in cognitive skill acquisition: A cognitive load perspective. *Educational Psychologist*, 38(1), 15–22. doi: 10.1207/S15326985EP3801_3
- Ries, E. (2011). *The lean startup: How today’s entrepreneurs use continuous innovation to create radically successful businesses*. Crown Business.
- Rother, M., & Shook, J. (1999). *Learning to see: Value-stream mapping to create value and eliminate muda*. Cambridge, MA: Lean Enterprise Institute.
- Schäfer, M., Nadi, S., Eghbali, A., & Tip, F. (2024). An empirical evaluation of using large language models for automated unit test generation. *IEEE Transactions on Software Engineering*, 50(1), 85–105. doi: 10.1109/TSE.2023.3334955
- Schmid, L., Hey, T., Armbruster, M., Corallo, S., Fuchß, D., Keim, J., ... Koziol, A. (2025). *Software architecture meets LLMs: A systematic literature review*. arXiv:2505.16697. doi: 10.48550/arXiv.2505.16697
- Sedano, T., Ralph, P., & Péraire, C. (2017). Software development waste. In *Proceedings of the 39th international conference on software engineering (icse)* (pp. 130–140). IEEE. doi: 10.1109/ICSE.2017.20
- Seddon, D. (n.d.). *Import linter*. <https://import-linter.readthedocs.io>. (Accessed September 2026)
- Shingo, S. (1986). *Zero quality control: Source inspection and the poka-yoke system*. Productivity Press.
- Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research*, 78(1), 153–189. doi: 10.3102/0034654307313795
- Siegmund, J. (2016). Program comprehension: Past, present, and future. In *Ieee 23rd international conference on software analysis, evolution, and reengineering (saner)* (pp. 13–20). IEEE. doi: 10.1109/SANER.2016.35
- Sobek II, D. K., & Smalley, A. (2008). *Understanding A3 thinking: A critical component of Toyota’s PDCA management system*. Productivity Press.
- Solis, C., & Wang, X. (2011). A study of the characteristics of behaviour driven development. In *37th euromicro conference on software engineering and advanced applications (seaa)* (pp. 383–387). IEEE. doi: 10.1109/SEAA.2011.76
- Spear, S., & Bowen, H. K. (1999, September). Decoding the DNA of the Toyota production system. *Harvard Business Review*, 77(5).
- Staats, B. R., Brunner, D. J., & Upton, D. M. (2011). Lean principles, learning, and knowledge work: Evidence from a software services provider. *Journal of Operations Management*, 29(5), 376–390. doi: 10.1016/j.jom.2010.11.005
- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285. doi: 10.1207/s15516709cog1202_4
- Sweller, J., & Cooper, G. A. (1985). The use of worked examples as a substitute for problem solving in learning algebra. *Cognition and Instruction*, 2(1), 59–89. doi: 10.1207/s1532690xc0201_3
- Terra, R., & Valente, M. T. (2009). A dependency constraint language to manage object-oriented software architectures. *Software: Practice and Experience*, 39(12), 1073–1094. doi: 10.1002/spe.931
- TNG Technology Consulting. (n.d.). *ArchUnit: Unit test your Java architecture*. <https://www.archunit.org>. (Accessed 2026)
- Tutors contributors. (2026a). *Tutors mono repo: an open-source course reader and learning platform*. <https://github.com/tutors-sdk/tutors-mono-repo>. (Read at commit 865d02f. Accessed 2026-09-30)
- Tutors contributors. (2026b). *Tutors release harness*. <https://github.com/tutors-sdk/tutors-release-harness>. (Version 1.15.0, read at commit c324525. Accessed 2026-09-30)
- Vaithilingam, P., Zhang, T., & Glassman, E. L. (2022). Expectation vs. experience: Evaluating the usability of code generation tools powered by large language models. In *Chi conference on human factors in computing systems, extended abstracts*. ACM. doi: 10.1145/3491101.3519665

- Wang, X., Conboy, K., & Cawley, O. (2012). “Leagile” software development: An experience report analysis of the application of lean approaches in agile software development. *Journal of Systems and Software*, 85(6), 1287–1299. doi: 10.1016/j.jss.2012.01.061
- Wang, X., Li, B., Song, Y., Xu, F. F., Tang, X., Zhuge, M., ... Neubig, G. (2025). OpenHands: An open platform for AI software developers as generalist agents. In *International conference on learning representations (iclr)*.
- Womack, J. P., & Jones, D. T. (1996). *Lean thinking: Banish waste and create wealth in your corporation*. Simon & Schuster.
- Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17(2), 89–100. doi: 10.1111/j.1469-7610.1976.tb00381.x
- Wynne, M., & Hellesøy, A. (2012). *The cucumber book: Behaviour-driven development for testers and developers*. Pragmatic Bookshelf.
- Yang, J., Jimenez, C. E., Wettig, A., Lieret, K., Yao, S., Narasimhan, K., & Press, O. (2024). SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in neural information processing systems (neurips)*.
- Zhang, Y., Ruan, H., Fan, Z., & Roychoudhury, A. (2024). AutoCodeRover: Autonomous program improvement. In *Proceedings of the 33rd acm sigsoft international symposium on software testing and analysis (issta)*. ACM. doi: 10.1145/3650212.3680384

A. Reading a receiver

Each receiver is self-contained. A reader interested in one language can start from its directory and read in this order.

1. AGENTS.md: the manifest. Verify command, placement table, conventions, requirement convention, feature recipe, and what needs a person first.
2. REQUIREMENTS.md: the ten rules in EARS, with ids and the pattern-to-layer table.
3. The worked slice: the domain entity, the service’s move-task method, the repository interface and its implementation, and the route or controller.
4. The scenarios: unit tests over the service with an in-memory repository, and API tests through the real composition root. Each cites the requirements it covers.
5. The rules: the architecture tests and the requirement check.

The evaluation scripts are in `evaluation/`: `measure.py` for size, `seed_violations.py` for seeding, and `agent_experiment/` for the pilot. The first paper’s results are kept alongside the current ones as `*-8c4a74a.json`.

Table 11 Where to find each part in each receiver.

Part	java/	python/	typescript/
Verify	mvn verify	pytest	npm test
Architecture rules	ArchitectureTest.java	test_architecture.py*	architecture.test.ts*
Requirement check	RequirementsTest.java	test_requirements.py*	requirements.test.ts*
Scenarios	service/, api/, events/ tests	tests/unit, tests/api	tests/unit, tests/api
Honest double	TaskRepositoryContract.java	tests/integration/	tests/integration/

* under tests/architecture/.

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